

Ben-Gurion University of the Negev
The Faculty of Natural Sciences
The Department of Physics

Di-Jet Topologies in Ultra Peripheral Collisions with the ATLAS Detector

PhD research proposal

Yftach Moyal

Under the supervision of Prof. Zvi Citron

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Signature of supervisor: _____
Signature of chairperson of the committee for graduate studies: _____

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Abstract

In recent years there has been an increased interest within the high energy and nuclear physics communities in the application of hadron colliders as photon-photon and photon-hadron colliders. The charge of the accelerated hadron generates a photon flux which allows for measurements that utilize photons as probes into QCD dynamics, nuclear structure, and saturation scale physics. For my research I propose to study these photon interactions using the ATLAS experiment at the Large Hadron Collider (LHC). In this document we will discuss photon interactions at the LHC, and the types of ultra peripheral (UPCs) ion-ion collisions which enable them. Then we will delve into open questions regarding specific collision geometries, such as photon emitter ambiguities in events without nuclear breakup, and how to address them in di-jet events. I will propose using machine learning models to differentiate between different UPC photoproduction sub-processes and how to use that framework to extract additional information regarding these events.

Contents

I Introduction **1**

1 The Standard Model and Quantum Chromodynamics **1**
1.1 QCD Jet . . . 2
1.2 Pomeron . . . 3

2 Photon Interactions **4**
2.1 Collision Kinematics . . . 4
2.2 Equivalent Photon Approximation . . . 4
2.3 Electromagnetic Dissociation . . . 5
2.4 Photon-Photon Interactions . . . 5
2.5 Photon-Hadron Interactions . . . 5
2.6 Photon-Pomeron Interactions . . . 7

3 Heavy Ion Collisions **7**
3.1 Heavy Ion Collisions . . . 7
3.2 Large Hadron Collider . . . 9
3.3 The ATLAS Experiment . . . 10
3.3.1 The Barrel Detector . . . 12
3.3.2 The Forward Region . . . 12
3.4 Jets . . . 13
3.4.1 Jet System Kinematic Variables . . . 14
3.5 Ultra Peripheral Collisions . . . 15

4 Deep Learning **15**
4.1 Neural network fundamentals . . . 16
4.2 Depth and hierarchical feature extraction . . . 16
4.3 Training via back propagation and stochastic gradient descent . . . 16
4.4 Regularization, generalization, and overfitting . . . 16
4.5 Architectural families . . . 17

5 Previous Analysis **17**

6 Current Analysis **18**

II Analysis and Results **19**

7 Event Selection **19**
7.1 Offline Selections . . . 19
7.1.1 Track and Topocluster Selections . . . 20
7.1.2 UPC→ Di-jet Selections and Cuts . . . 20
7.1.3 Non Physics Background . . . 21
7.2 Selection Summary . . . 23

8 Estimating Photon Direction **23**
8.1 Estimating Photon Direction in UPC dijet events 23

9 Analysis Through Event Topology **27**
9.1 Classification model based on event topology 27
9.2 Data preparation 27
9.3 Loss function 29
9.4 Training the model 30

10 Results **30**
10.1 Photon Direction 30
10.2 UPC Process Discrimination 33
10.3 Sample Discrimination 33
10.3.1 Multi Process Discrimination 33
10.3.2 Binary Process Discrimination 34

III Follow up research **37**

List of Figures

1 The table describing the standard model. Quarks outlined in purple and leptons in green [1]. 2
2 Illustration of a flux tube breaking, generating new quarks and anti-quarks in the process [3]. 3
3 (a) Quark fragmentation to hadrons induced by confinement forces; (b) quark/gluon cascades at high energies in QCD [4]. 3
4 Product of the spectrum of equivalent photons and total photonuclear cross section for 10.6 A GeV gold nuclei (dotted curve) and 158 A GeV lead nuclei (solid curve) in the interactions with gold and lead targets, respectively. The regions of GR excitation, quasi-deuteron absorption,and Δ-isobar excitation are shown. For comparison, a product of the photon spectrum at an energy of 158 A GeV and the partial cross sections of photonuclear reactions with ^{83}Bi (dashed curve) is shown [8]. 6
5 Contributions to hard γp interactions: (a) direct, (b) VMD, and (c) anomalous. Only the basic graphs are illustrated; additional partonic activity is allowed in either case. The presence of spectator jets has been indicated by dashed lines, while full lines show partons that (may) give rise to high-p_T jets [10]. 7
6 Typical leading-order Feynman graphs for diffractive di-jet photoproduction in UPCs of hadrons A and B. Graphs a and b correspond to the direct and resolved photon contributions,respectively [11]. 8
7 Illustration of the collision geometry as function of impact parameter b. The blue region represents the overlapping region in which collision participants interact hadronically, while nulceons in the red region are spectators to the event. 9

8 Correlation between the total energy measured in the two sides of the ZDC and the total transverse energy measured in the ATLAS forward calorimeter (FCal). The faint second band above the main band results from events having in-time pileup [12]. 9
9 A schema of the LHC complex with the injectors and different accelerator rings. Different facilities are color coded and dated [14]. 10
10 Cut-away view of the ATLAS ID, which is designed to provide a high-precision reconstruction of charged-particle trajectories. The ID covers the pseudo-rapidity range $|\eta| < 2.5$ and has full coverage in ϕ. It consists of a silicon Pixel detector at the innermost radii surrounded by a silicon microstrip detector (SCT) and a straw-tube detector, the TRT, that combines continuous tracking capabilities with particle identification based on transition radiation [15]. 11
11 Cut-away view of the ATLAS calorimeter system that measures the energies and positions of charged and neutral particles through interleaved absorber and active layers out to $|\eta| < 4.9$. LAr is used as the active medium for the electromagnetic calorimeters and all of the endcap and forward calorimeters; scintillating Tiles are used for hadron calorimetry in the central region.[15] 11
12 The distribution of energies, expressed in terms of number of neutrons, in the A and C sides of the ZDC for events having at least 1 neutron on each side. The 1, 2, 3, and 4 neutron peaks can be seen as well as a large enhancement at 40 neutrons arising from Pb+Pb collision geometry and the physics of (charged) nuclear cluster formation which reduces the neutron yield in peripheral collisions [12]. 13
13 Event display of a monojet event and its associated tracks and clusters in ATLAS [16]. Tracks are seen as orange lines and calorimeter energy is seen as green (EM) and yellow (HAD) boxes, with size proportional to energy deposit. 14
14 Representation of the cuts (in red lines) used to remove splash events. As expected, there is much higher contribution of splash events in the 0nXn (bottom two plots) sample due to the probability of neutral fragments reaching the ZDCs. 22
15 Representation of the cuts (in red lines) used to remove OOT pileup. 23
16 Plot of calibrated EM Topojets time with Gaussian fit. 5σ bands are shown in magenta to describe the cuts needed to reduce beam halo muons. 25
17 y_{jets} distribution for different MC samples. Nucleus emitting the probing γ is traveling towards forward rapidity. The $\gamma\mathbb{P}$ contribution includes 2 peaks: one closer to mid-rapidity, which stems from events where the nucleus underwent coherent diffraction, and one closer to backward rapidity, where the diffraction was stemmed from incoherent interaction. 25
18 Event Display of a single γA event with the nucleus emitting the photon going towards positive η. The event is overlaid with the edge gaps and the particle bands. Edge gaps are in gray,A side band in green and C side band in red, with the reference point being the mean di-jet position in η. This specific example also shows the edge case in which detector acceptance can change the answer. . 26

19 Diagram of the ML model. The track and topocluster maps receive the same treatment by two different instances of a ResNet 50 variant network, before being concatenated with tabular event data. Each of the ResNet blocks can generate activation maps of the filters associated with it. 28
20 η-ϕ map of a single event, where both tracks and clusters are present. (a) Event map prior to rotation. (b) Post rotation. 29
21 Confusion matrices for the training and testing samples for photon side discrimination. The plots show slight overfitting, but unbiased prediction on both samples. 31
22 Accuracy plots for γA (left) and $\gamma\mathbb{P}$ (right) binned in H_T. Different colors assigned to the different methods used for the calculation. 32
23 Accuracy plots for γA (left) and $\gamma\mathbb{P}$ (right) binned in y_{jets}. Different colors assigned to the different methods used for the calculation. 32
24 Accuracy plots for 0nXn binned in H_T (left) and y_{jets} (right). "Truth" information defined by ZDC X direction. 33
25 Confusion matrices for the training and testing samples for multi-class classification. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions. There is some block structure showing separation of $\gamma\gamma$ from γA , $\gamma\mathbb{P}$ contributions, while there is some overlap between $\gamma\mathbb{P}$ and $\gamma\gamma$RR. 34
26 Confusion matrices for the training and testing samples for binary $\gamma\gamma$, γA where all $\gamma\gamma$ processes were labeled as one. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions. . . . 35
27 Confusion matrices for the training and testing samples for γA and $\gamma\mathbb{P}$ processes. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions. 35
28 Confusion matrices for the training and testing samples for $\gamma\gamma$ and $\gamma\mathbb{P}$ processes. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions. 36
29 Diagram of the decision process for final sample classification based on the results of the different models. 36
30 Plots of y_{jets} for 0nXn (Left) and 0n0n (Right), separating out the different contributions according to the ML models. Events were selected based on the photon direction model such that the nucleus emitting the photon goes in positive rapidity. 37

Part I
Introduction

This chapter will discuss heavy ion (HI) collisions, and the ability to measure them using the ATLAS detector. It will then focus on ultra peripheral collisions with emphasis on the theoretical frameworks and standard model interactions associated with them. Finally, we will introduce basic concepts of machine learning and deep learning that will be used later to analyze specific UPC processes.

1 The Standard Model and Quantum Chromodynamics

The standard model of particle physics is currently the most comprehensive and experimentally validated framework describing the fundamental particles and their interactions. It successfully unifies three of the four fundamental forces: electromagnetism; the weak nuclear force; and the strong nuclear force, into a single theoretical structure. Within this model, all known elementary particles are categorized into two primary classes: fermions and bosons. Fermions, which include quarks and leptons, are the building blocks of matter. Quarks combine to form mesons (e.g J/ψ) and baryons (e.g protons and neutrons), while leptons are elementary particles which include electrons, muons, tau and their associated flavor neutrinos. In contrast, bosons are force carriers that mediate interactions between fermions, transmitting the fundamental forces through the exchange of virtual particles. All the particles are listed in the illustration in Fig. 1.

Among the bosons, the photon serves as the mediator of the electromagnetic interaction, responsible for transmitting electric and magnetic forces between particles charged under it. The weak interaction, which governs processes like beta decay and neutrino interactions, is mediated by the W^+, W^-, and Z^0 bosons. In contrast, the strong interaction, which binds quarks together within protons, neutrons (and other hadrons) and also holds nuclei together, is mediated by gluons. Contrary to the photon, gluons are charged under the strong force. This causes a phenomenon called self-interaction, leading to extremely complex, non-perturbative structures. This includes the formation of glueballs, hypothetical particles composed entirely of gluons and gluon ladders. Gluon ladders, specifically, will be expanded upon in later section. Unlike electromagnetism, where the force is mediated by photons that do not interact with each other, the strong interaction is mediated by gluons, which are massless bosons, charged under the strong force, with charge which is referred to as color. Contrary to the electric charge, color charge has 6 unique elements, labeled as red, green, blue and their counterparts (i.e - anti-red). The strong interaction is responsible for binding the nucleons, which are comprised of thee generations of quarks (anti-quarks) and gluons together, such as proton (nominally uud) and neutrons (nominally udd).

As a result of the symmetry and non-Abelian nature of QCD, there cannot be free particles with color charge. Instead, all observed particles are color-singlets. Contrary to electromagnetism, one of the proposed potential of the strong interaction, is linear with distance at large distances (Cornell potential [2]):

$$V(r) = -\frac{4}{3}\frac{\alpha_s}{r} + \sigma r + Const\,, \tag{1}$$

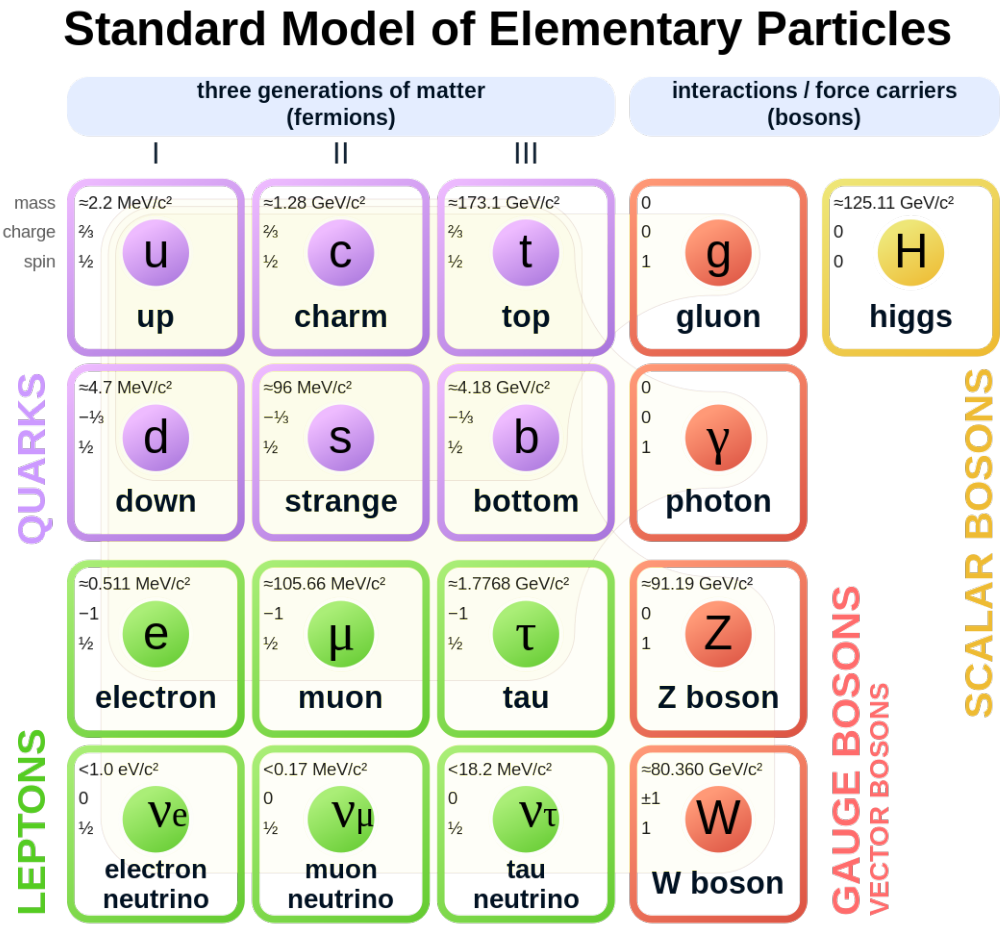


Figure 1: The table describing the standard model. Quarks outlined in purple and leptons in green [1].

where α_s is the strong coupling and σ is called the string tension, related to the flux tubes connecting color-charged partons. Due to this potential, as color charged particles drift apart, the potential energy increases, until eventually it is preferable to split the flux tube between the particles and generate new color charged particles in place, as illustrated in Fig. 2.

1.1 QCD Jet

As discussed previously, quarks and gluons are bound by the strong interaction and will generate new color partons when pulled apart. Following that, free quarks and gluons in a highly boosted frame of reference will generate a shower of color partons through a bremsstrahlung cascade, which will subsequently undergo hadronization (see Fig. 3) and will result in highly collimated clusters of hadrons, leptons and photons [4]. Additional information regarding the measurable quantity known as jet will be explained in section 3.4.

The process of hadronization is a topic of active research. Currently research into hadronization uses various models and simulations, as perturbative QCD is not applicable at these ener-

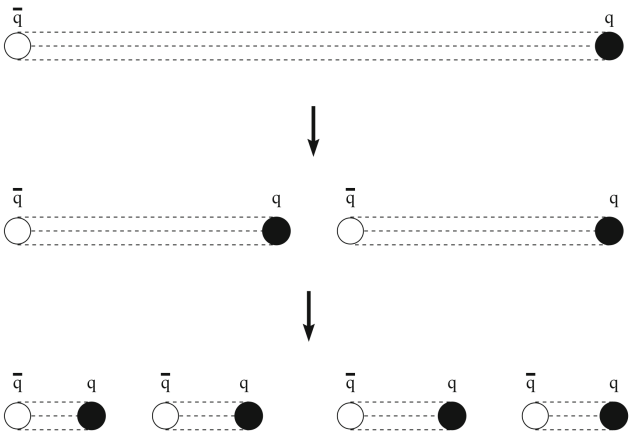


Figure 2: Illustration of a flux tube breaking, generating new quarks and anti-quarks in the process [3].

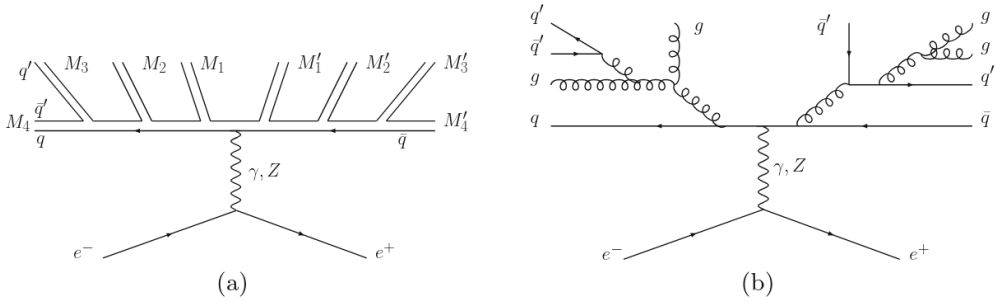


Figure 3: (a) Quark fragmentation to hadrons induced by confinement forces; (b) quark/gluon cascades at high energies in QCD [4].

gies.

1.2 Pomeron

The pomeron (usually denoted $\mathbb{P}$) is a theoretical, effective object in high-energy scattering processes. It is not a fundamental particle but rather a composite structure that mediates diffraction in scattering. The properties of the pomeron include no color charge, no electric charge and zero spin. The pomeron properties lent themselves to describing infinite gluon ladders. These are usually referred to as a chain of strong interactions between quarks and gluons, including gluon self-interactions. Specifically, gluon ladders are a pictorial and mathematical description of the propagation of gluons in high-energy scattering events. Gluon ladders refer to sequences of virtual gluon exchanges that form a 'ladder-like' structure in Feynman diagrams, where each rung of the ladder represents a gluon pair (gluon-gluon interaction) that mediates the interaction between two high-energy partons (quarks or gluons) in a scattering process. These diagrams are particularly prominent in the regime of large momentum transfer and are central to understanding the behavior of partons in deep inelastic scattering (DIS) and high-energy hadronic collisions. These infinite ladders are color-singlet, as required by the

pomeron description, and can still mediate diffraction via interaction with the gluons within the ladder. Specifically, protons and neutrons can interact with photons by pomeron exchange. The pomeron then acts as the mediator enabling momentum transfer between the nucleon and the photon.

2 Photon Interactions

It is important to establish the context in which we will want to study photons. One of the novel ways researchers use the LHC in the last few years is as a Large Photon Collider. This is done by utilizing both pp, pA and A + A ultra peripheral collisions as sources of photons due to high acceleration of charged particles, combined with lack of hadronic interactions due to geometry constraints. As a result, one can measure extremely high energy photon-photon and photon-matter collisions, making it a very exciting probe for a variety of questions both in nuclear and high energy physics [5, 6, 7]. Therefore, the following section will describe photon interactions under the lens of "photon colliders".

2.1 Collision Kinematics

In order to establish the premise for the interaction, this subsection will discuss several key terms related to particle collision kinematics. Firstly, we shall define $-Q^2$ as the square of the momentum transfer between incoming particle and the target particle. In case of a photon with four-momentum q^μ, one would get $Q^2 = -q^2$. Another important quantity is Bjorken x , or x for short, which is defined as:

$$x = \frac{Q^2}{2pq}\,, \tag{2}$$

where p is the momentum of the target. Bjorken x can be thought of as the fraction of momentum carried by the quark which interacted with the photon.

2.2 Equivalent Photon Approximation

The Equivalent Photon Approximation (EPA) is a framework in high-energy nuclear physics for modeling electromagnetic interactions in UPCs, where the nuclei do not undergo hadronic interaction. The core idea of the EPA is to represent the time-varying Coulomb field of a relativistically moving nucleus as a flux of virtual or real photons with a broad, continuous energy spectrum. This approach, first introduced by Fermi, and later formalized by Weizsäcker and Williams [8] treats the field of an accelerated nucleus as an effective photon flux, allowing one to describe the interaction of a target nucleus with this "photon field" in a manner analogous to photon absorption by a stationary target.

The formalism imposes that the transverse photon flux produced by the relativistic nucleus can be described by equivalent, linearly polarized virtual photons, where the number of photons is given by [9]:

$$N(k,b) = \frac{\alpha Z^2 k^2}{\pi^2 \gamma^2 \hbar^2 \beta^2} \left(K_1^2(x) + \frac{K_0^2(x)}{\gamma^2} \right)\,, \tag{3}$$

where Z is the atomic number, α is the fine structure constant, γ and β are the Lorentz boost and velocity respectively. K_0 and K_1 are the Bessel functions and $x = kb/\ \beta\gamma\hbar c$, where b is the impact parameter. The k_T spectrum of these photons is given by:

$$\frac{d^2N}{d^2k_T} = \frac{\alpha Z^2}{\pi^2}\frac{F(k_T^2 + k_z^2/\gamma^2)}{(k_T^2 + k_z^2/\gamma^2)^2}k_T^2\,, \tag{4}$$

where this equation was obtained by integrating over all possible values of b. Requiring coherent photon emission from the nucleus (i.e dipole-dominant contribution $Q^2R^2 \leq 1$) implies that photons emitted from nuclear target are quasi-real and not virtual as usually seen when assessing e^+e^- colliders. This framework allows us to discuss the interactions of the photon flux from one nucleus with another in terms of equivalent photons that can be interpreted in Feynman diagrams.

2.3 Electromagnetic Dissociation

Electromagnetic Dissociation (EMD) is a process in which the nucleus or nucleon is excited by a photon, and emits nucleons as it comes down to the ground state. This effect relies on high-energy photon impinging upon the target nucleus, and therefore its' cross section increases with $\sqrt{s_{NN}}$. The exact process that takes place depends on the incoming photon energy, as can be seen in Fig. 4. For heavy nuclei such as Au and Pb, up to ($\sim$10 GeV) the dominant contribution is known to be from the Giant Dipole Resonance (GDR), which is a process in which the excitation of the nucleus causes the protons to move in unison with respect to the neutrons, generating a dipole within the nucleus itself. De-excitation of GDR is usually accompanied by neutron emission, and it is one of the mechanisms that generate spectator neutrons which are read by the Zero Degree Calorimeter (see section 3.3).

As the photon energy increases, other channels become available due to the higher resolution provided by a shorter light pulse, and the process of interaction with quasi-deuteron (np pair) is unlocked. Eventually, with photon energy of $\gtrsim$ 100 MeV the channel for interaction with individual nucleons is possible. Higher probe energies will allow the photon to interact with the nucleon constituents, the quarks and gluons, and will be the topic of the next sections.

2.4 Photon-Photon Interactions

As mentioned above, photons do not interact with each other in first order. Their interaction comes about through virtual fermions in higher-order loop diagrams within the framework of quantum electrodynamics (QED). In QED, virtual particle-antiparticle pairs such as electron-positron pairs can be created from the vacuum due to excitations of the fermion fields by the photon. These virtual loops allow for a non-trivial interaction between two real photons, effectively enabling a photon–photon scattering process. This interaction arises through a second-order Feynman diagram where one photon emits a virtual electron-positron pair, which is then reabsorbed by the second photon.

2.5 Photon-Hadron Interactions

In addition, theory suggests real (or quasi-real) photons can also interact with hadrons in two different ways: Through Vector Meson Dominance (VMD) fluctuations or through QCD

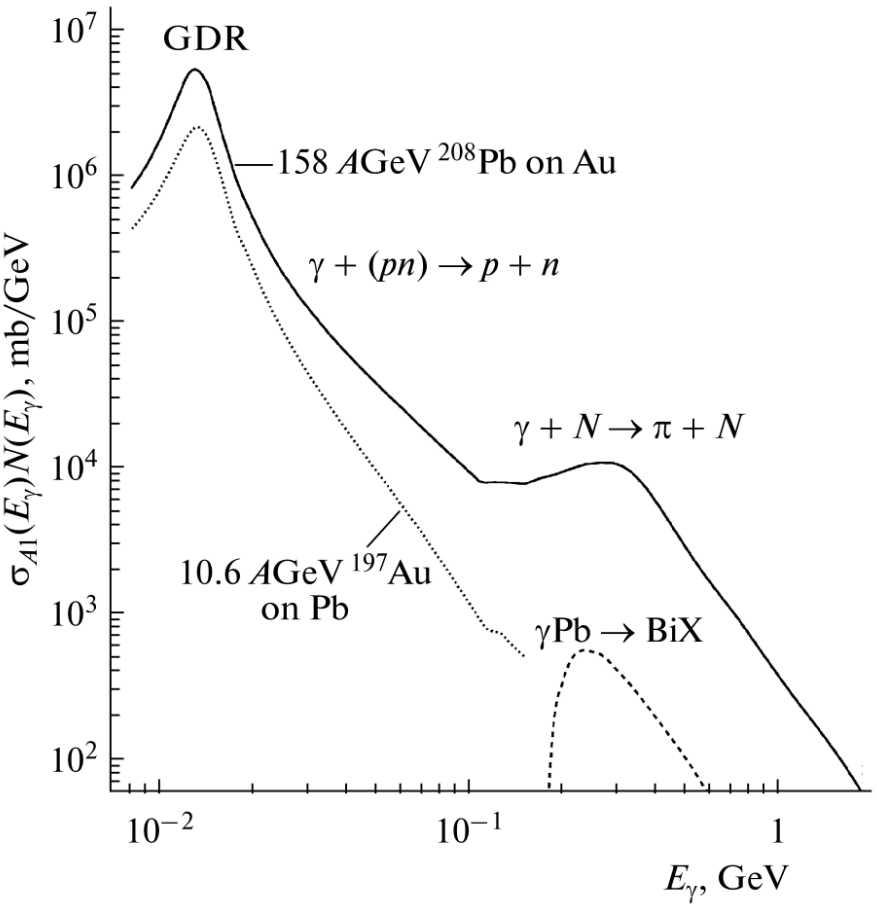


Figure 4: Product of the spectrum of equivalent photons and total photonuclear cross section for 10.6 A GeV gold nuclei (dotted curve) and 158 A GeV lead nuclei (solid curve) in the interactions with gold and lead targets, respectively. The regions of GR excitation, quasi-deuteron absorption,and Δ-isobar excitation are shown. For comparison, a product of the photon spectrum at an energy of 158 A GeV and the partial cross sections of photonuclear reactions with ^{83}Bi (dashed curve) is shown [8].

fluctuations to $q\bar{q}$ pairs. To discuss such descriptions, Schuler and Sjöstrand [10] describe the wave function of the photon as a superposition:

$$|\gamma\rangle = \sqrt{Z_3}\,|\gamma_B\rangle + \sum_{V=\rho^0,\omega,\phi} \frac{e}{f_V}\,|V\rangle + \frac{e}{f_{q\bar{q}}}\,|q\bar{q}\rangle \ , \tag{5}$$

where Z_3 is $1-\Sigma_{V=\rho^0,\omega,\phi}\left(\frac{e}{f_V}\right)^2-\left(\frac{e}{f_{q\bar{q}}}\right)^2$, $|\gamma_B\rangle$ is the "bare" photon, or the photon component as point particle we associate with EM theory and QED, and therefore cannot interact at all with hadrons. $|V\rangle$ is the vector meson component and $|q\bar{q}\rangle$ is the hadron component. The summation on ρ^0, ω, ϕ is due to them being the only hadronic constituents of the photon under simple VMD. VMD description however, breaks down in small Q^2 and small x, at which the final expression of the photon $|q\bar{q}\rangle$ takes effect and become the contributor to photon-parton interactions. In the proposal, we shall refer to photon interactions in which the latter is in

effect as "resolved" photon interaction, to denote the dissociation into $q\bar{q}$ pair which will then interact with the target.

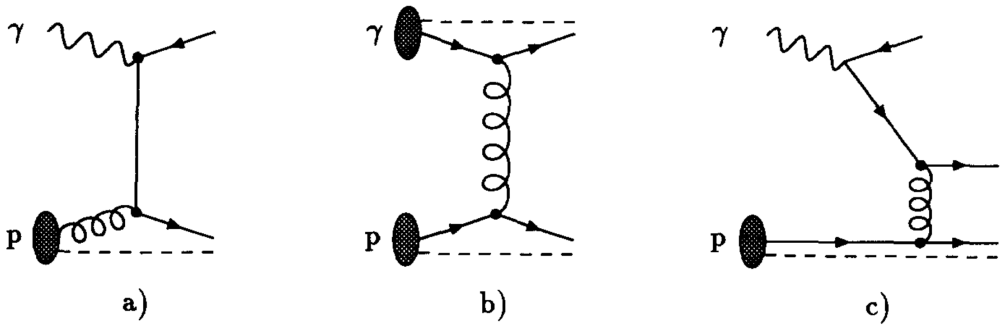


Figure 5: Contributions to hard γp interactions: (a) direct, (b) VMD, and (c) anomalous. Only the basic graphs are illustrated; additional partonic activity is allowed in either case. The presence of spectator jets has been indicated by dashed lines, while full lines show partons that (may) give rise to high-p_T jets [10].

2.6 Photon-Pomeron Interactions

As mentioned in 1.2, the pomeron photon interactions are a form of effective theory to describe collisions in which the photon interacts with a charge, color neutral complex QCD object during diffraction events. An example for such a process can be seen in Fig. 6. Here, as described above, the photon contribution is through a resolved state by dissociation into a $q\bar{q}$ pair, which then interacts with the pomeron by means of gluon exchange. It is important to notice that through this exchange there will be additional particle production from the pomeron, denoted in X.

There are two distinct types of photon-pomeron interaction, which are often called coherent and incoherent interactions and stem from the pomeron itself. In coherent interaction, the pomeron mediates an exchange from the entire nucleus, while in incoherent interaction the pomeron is associated with a single nucleon. Incoherent interaction then leaves the probed nucleus either in an excited state or causing it to break up. This is in contrast to coherent interaction in which the nucleus will most likely remain intact.

3 Heavy Ion Collisions

3.1 Heavy Ion Collisions

At the Large Hadron Collider (LHC), experiments are run to probe subatomic interactions. These experiments are carried out by colliding beams of particles at interaction points (IPs), in which detector complexes are built in order to measure the outgoing particles of the collisions. The beams are being controlled by a complex array of superconducting electromagnets which adjust the beams to collide at the aforementioned IPs. Through most of the year, the LHC is run in pp mode - i.e colliding bunches of protons with other protons. However, there are

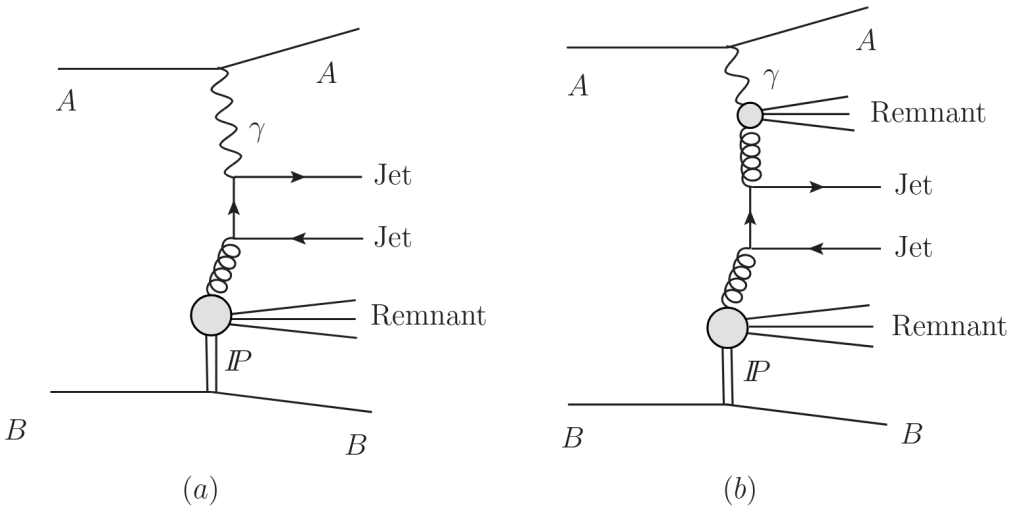


Figure 6: Typical leading-order Feynman graphs for diffractive di-jet photoproduction in UPCs of hadrons A and B. Graphs a and b correspond to the direct and resolved photon contributions,respectively [11].

specific programs in which heavy ions are collided. For the rest of the thesis we will concern ourselves with Pb + Pb collisions, in which both beams are comprised of Pb nuclei.

These Pb nuclei can interact in the IP in several ways, depending on the initial event geometry. This initial geometry is often referred to as the centrality of the event. We usually think of nuclei - nuclei interaction in terms of the nucleon pairs which interacted with each other (e.g proton-proton or proton-neutron). we call the number of pairs which had a head-on interaction as the number of participants in the collision. The nucleons which did not participate in the collision are called spectators. The nomenclature for these types of events is as follows:

- UCC (Ultra Central Collisions)
- CC (Central Collisions)
- PC (Peripheral Collisions)
- UPC (Ultra Peripheral Collisions)

As centrality varies from collision to collision, one has to extract it from the final state of each collision. One way to do that is by looking at the outcome of the collision and searching for remnants of the original nuclei. As established prior, these remnants are spectators to the collision, and through a straightforward analogue to the centrality as the overlap of the two nuclei, the spectators are a probe into the centrality of the event.

However, it is not a one-to-one translation. As a result of the beam magnets, remnants that include charge (protons and heavier nuclei such as ^{3}He, etc.) are bent away by the magnets and will not reach a detector, and thus information is lost. The only remnants not affected by the magnets are neutrons, which can be counted using a detector sitting along the beam line (section 3.3.2). The number of spectator neutrons can then be mapped to two different regions

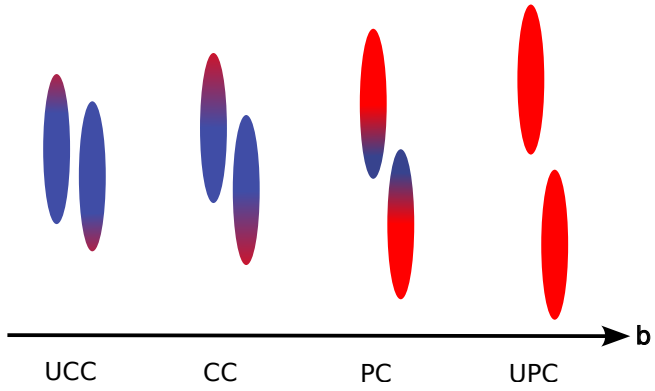


Figure 7: Illustration of the collision geometry as function of impact parameter b. The blue region represents the overlapping region in which collision participants interact hadronically, while nulceons in the red region are spectators to the event.

in centrality, which is usually demonstrated by forward calorimeter energy as in Fig. 8. In this case, discrimination on the centrality is done via the forward energy, as a selection independent from the number of tracks, which is used in most analysis.

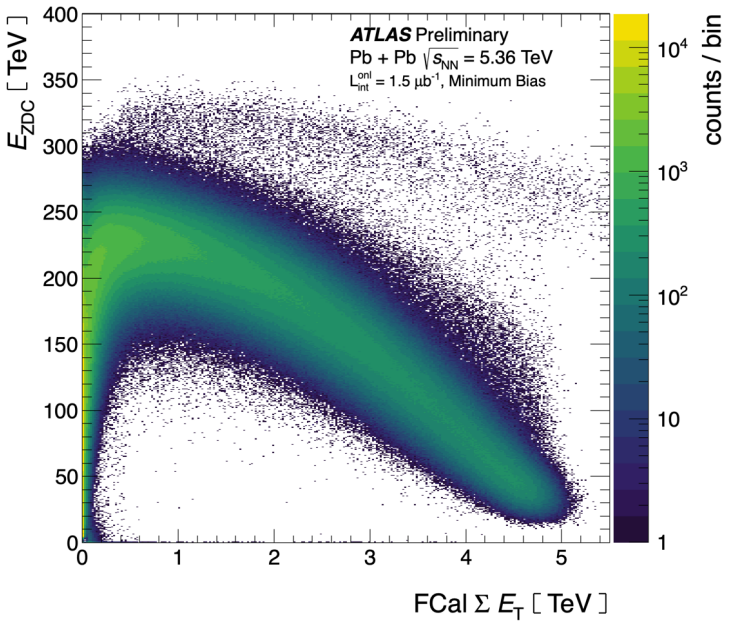


Figure 8: Correlation between the total energy measured in the two sides of the ZDC and the total transverse energy measured in the ATLAS forward calorimeter (FCal). The faint second band above the main band results from events having in-time pileup [12].

3.2 Large Hadron Collider

The Large Hadron Collider (LHC) is currently the biggest collider in the world for hadrons. It facilitates measurements of pp and A+A collisions within 4 major experiments: ATLAS,

ALICE, CMS and LHCb. The LHC had its major success after its first running at 2012 when the Higgs boson was discovered[13]. The collider is built out of multiple acceleration rings, which were used previously in LEP, etc. to accelerate protons and ions to TeV scale energies. A schematic of the accelerator complex can be seen in Fig. 9.

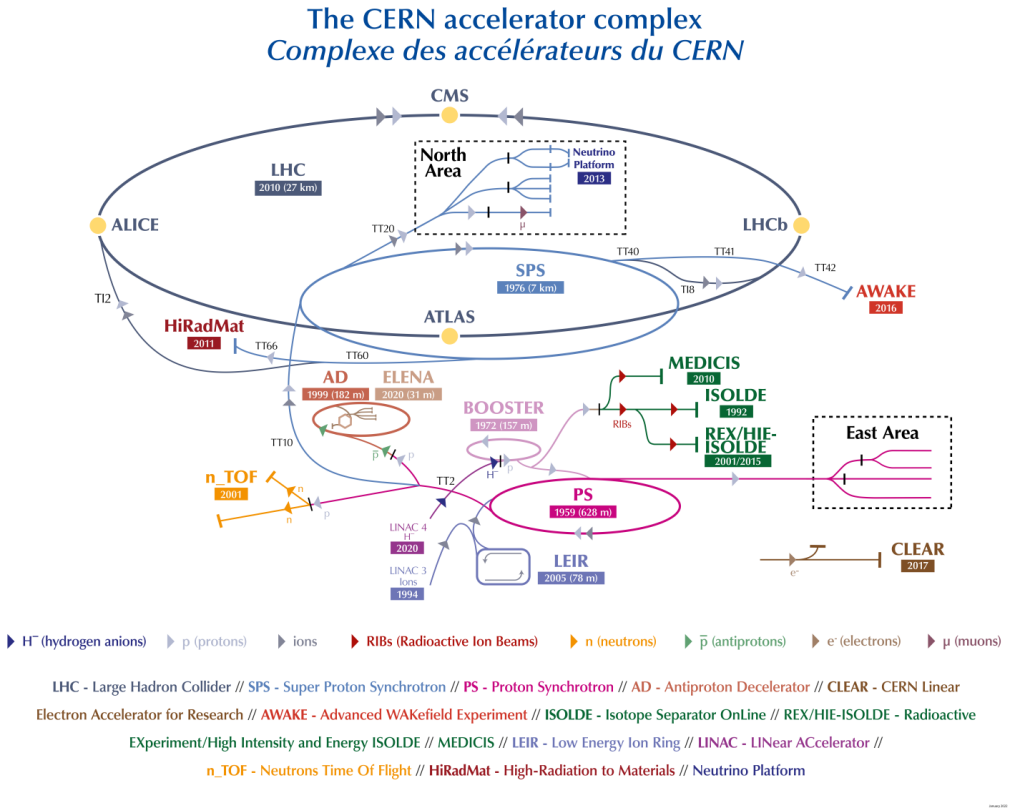


Figure 9: A schema of the LHC complex with the injectors and different accelerator rings. Different facilities are color coded and dated [14].

Heavy ion operations at the LHC are typically done once a year, and are usually designed around accelerating Pb ions, although other ions such as O, Ne and Xe have been run as special runs as well.

3.3 The ATLAS Experiment

The ATLAS (A Toroidal LHC Apparatus) Detector is one of the four major experiments in the LHC, situated at IP1. The detector is a general purpose detector for High Energy Physics (HEP), comprised of a main barrel, endcaps, and forward detectors. The main barrel has the inner detector (ID) tracking layers (pixel, SCT, TRT), calorimeter layers, and dedicated muon detection layers.

The lab coordinate system of ATLAS has its origin in the IP, with the z axis pointing along the beam axis towards IP2, and the x axis pointing towards the center of the LHC ring. Usually, the coordinate system used in the experiment is not the Cartesian system, but a somewhat Lorentz invariant system defined as follows: Taking a cylindrical coordinate along

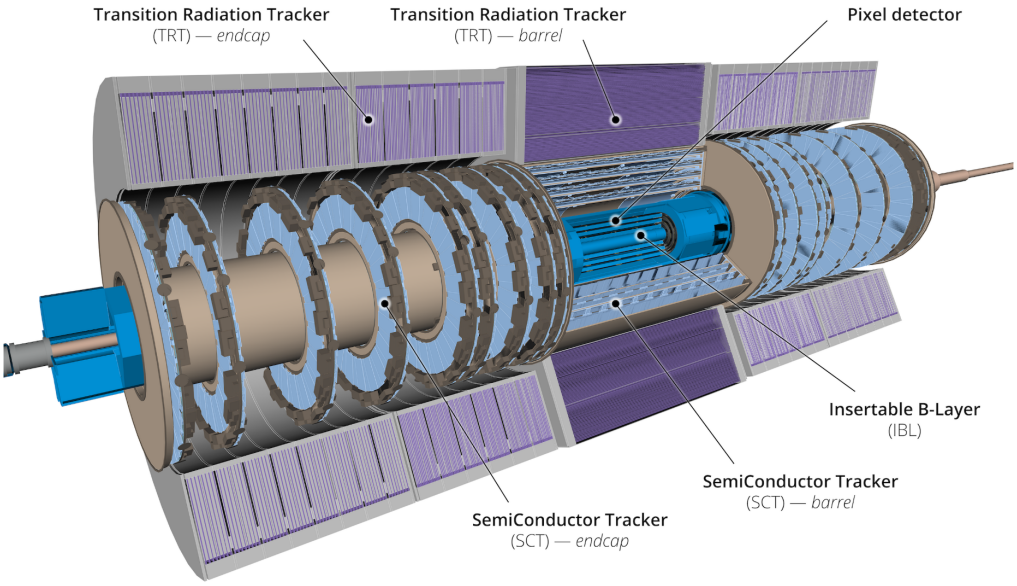


Figure 10: Cut-away view of the ATLAS ID, which is designed to provide a high-precision reconstruction of charged-particle trajectories. The ID covers the pseudo-rapidity range $|\eta| < 2.5$ and has full coverage in ϕ. It consists of a silicon Pixel detector at the innermost radii surrounded by a silicon microstrip detector (SCT) and a straw-tube detector, the TRT, that combines continuous tracking capabilities with particle identification based on transition radiation [15].

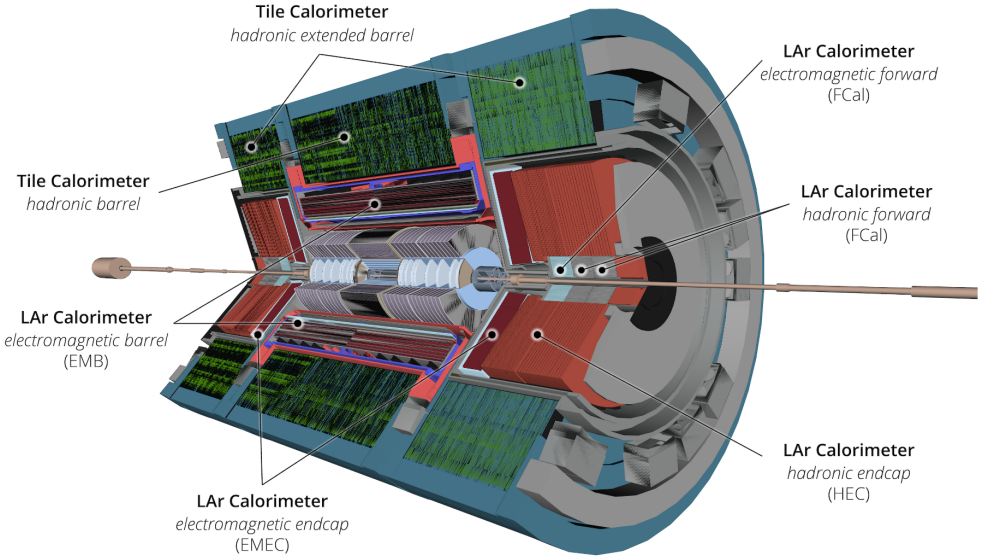


Figure 11: Cut-away view of the ATLAS calorimeter system that measures the energies and positions of charged and neutral particles through interleaved absorber and active layers out to $|\eta| < 4.9$. LAr is used as the active medium for the electromagnetic calorimeters and all of the endcap and forward calorimeters; scintillating Tiles are used for hadron calorimetry in the central region.[15]

the beam axis (z is pointing in the same direction as the Cartesian case), we define the pseudo rapidity as the following:

$$\eta = -ln\left(tan\left(\frac{\theta}{2}\right)\right) = \frac{1}{2}ln\left(\frac{|\mathbf{P}| + P_z}{|\mathbf{P}| - P_z}\right). \tag{6}$$

Therefore, the difference $\Delta\eta$ is Lorentz-invariant, which makes it easy to perform calculations.

3.3.1 The Barrel Detector

As mentioned, the barrel detector wraps around the IP, and therefore is able to provide an almost full geometric coverage for the outgoing particles in the event. It is convenient to separate it into three different sections based on function:

Tracking The tracking layers are responsible for reconstructing charged particle trajectories close to the IP. This is done by an array of silicon pixels (pixel detector), silicon strips (Semiconductor Tracker (SCT)) and argon mixture drift tubes (Transition Radiation Tracker (TRT)). These detectors sit inside a 2T solenoid magnet, bending the particle trajectory of charged particles, allowing for momenta reconstruction. It is essential that the particles interacting with the tracking layers leave minimal energy deposit in it, in order to not affect the momentum and energy of the particle.

Calorimetry The calorimetry layers are designed to stop particles inside the active area, while converting the energy into a signal. The first layer past the tracker is the electromagnetic calorimeter (also known as the LAr calorimeter), followed by the hadronic calorimeter (Tile Calorimeter). The principle of operation is to use high Z material to create showers of particles inside the LAr. Then the charged particles from the showers produce current proportional to the energy of the initial particle, which is then read out. The Tile Calorimeter works in the same way, but instead of LAr, plastic scintillators are used to measure the energy of showers not stopped in the LAr. The Tile Calorimeter has a gap at $\eta \approx \pm 1.7$ due to the geometry and construction of the scintillators, and has to be accounted for in the analysis.

Muon Detectors The muon detectors are a set of spectrometers outside the calorimeters. Muons have a small interaction cross-section and above 2 GeV they will not be stopped in the ID or calorimetry, whereas other observable particles typically are. Thus the particle measured in the muon detectors are largely identified as muons. These sub-detectors are able to provide tracking as well as timing information regarding the muons.

3.3.2 The Forward Region

Zero Degree Calorimeter (ZDC) The ZDC is a detector placed 140m away from the IP in both directions, and sits inside the Target Absorber for Neutrals (TAN). Following its name, it measures particles moving at around zero degrees with respect to the beam line, which, given the beam line magnets, will be neutral particles, specifically neutrons and photons. The ZDC is a calorimeter, built using alternating layers of tungsten (W) and fused silica/quartz rods. When a particle interacts with the tungsten, it creates a shower

of particles,some of them charged. the charged particles propagate through the rods, and due to their momenta and index of refraction, produce Cherenkov light inside the rods. Through total internal reflection, the Cherenkov light is then propagated through the rods to photomultiplier tubes (PMTs) which convert the photons into electric signals which are then read out. Although the ZDC can measure photons, its main role is in counting spectator neutrons. As the energy of a single spectator neutron is the beam energy, one can use the energy deposit in the ZDC to reconstruct how many spectator neutrons were present on each side of the collision in the event.

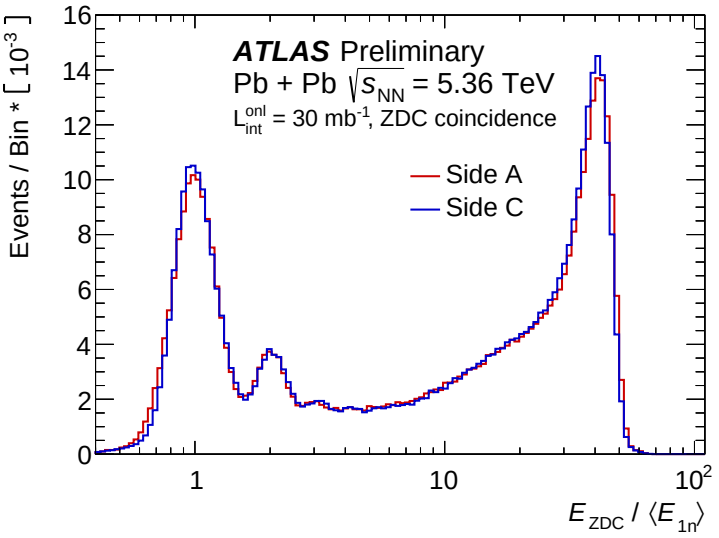


Figure 12: The distribution of energies, expressed in terms of number of neutrons, in the A and C sides of the ZDC for events having at least 1 neutron on each side. The 1, 2, 3, and 4 neutron peaks can be seen as well as a large enhancement at 40 neutrons arising from Pb+Pb collision geometry and the physics of (charged) nuclear cluster formation which reduces the neutron yield in peripheral collisions [12].

3.4 Jets

Jets, in the most abstract sense are a collection of multiple detector measurements, usually energy clusters but tracks may be used as well, confined to a predetermined radius within the detector $\eta - \phi$ plane. The definition of jet therefore depends on the specific jet algorithm in question and the choice of parameters when applying the algorithm to data. Jets are used to account for complexities which arise from the interplay between quarks and gluons when hadronization occurs, as described above. To be more precise, jets are proxies to the collimated partons produced in the interaction. In general, there are a few types of jets used in ATLAS analysis:

EM Topo Jets Jets created only from calorimeter energy deposits. These jets have timing information from the calorimeter, as well as accounting for photons as constituents of the jets.

Particle Flow Jets These jets are created using a more complex algorithm which also adds track information to the forming of the jet. However, the resulting time resolution of these jets is poor compared to topojets, and as such operations required on jet timing are worse off with these kinds of jets.

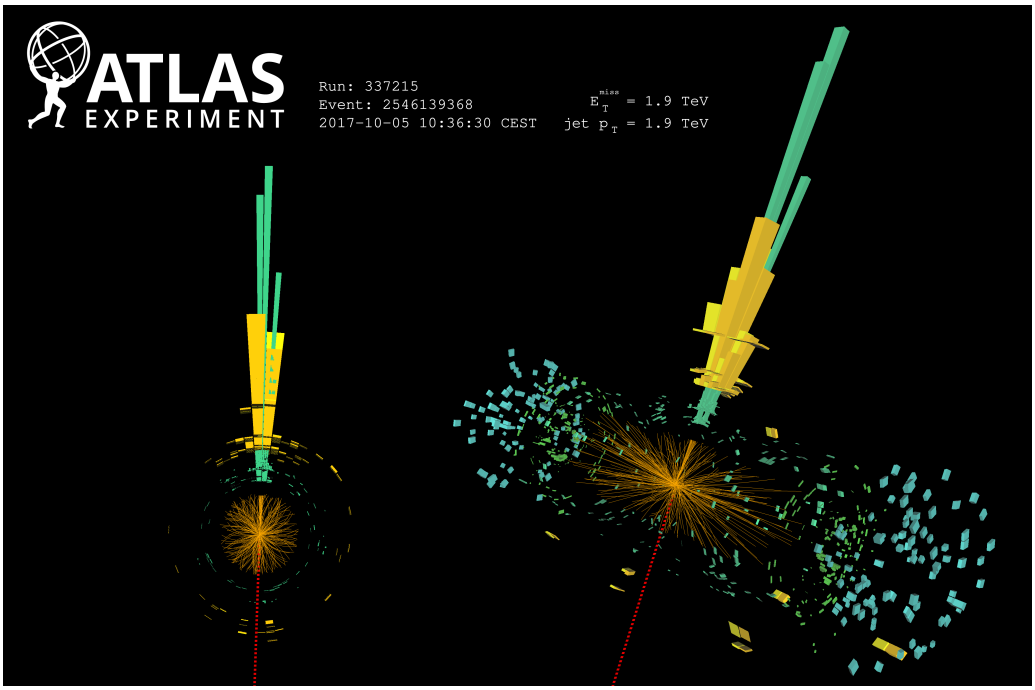


Figure 13: Event display of a monojet event and its associated tracks and clusters in ATLAS [16]. Tracks are seen as orange lines and calorimeter energy is seen as green (EM) and yellow (HAD) boxes, with size proportional to energy deposit.

The main algorithm currently used to define jets is the anti-k_T algorithm [17]. the main idea behind the algorithm is to combine cluster deposits based on the negative power of the energy scale (hence anti-k_T), which makes the clusters generated have defined, cone-like boundaries. The process runs iteratively on clusters and pseudo-clusters (clusters which were already formed by previous iterations) up until a defined radius for the cone is reached (one of the more prominent selections is R=0.4). Then the constituents of the jet are charged tracks and clusters in which $\Delta R \leq 0.4$, where

$$\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2}\,, \tag{7}$$

with $\Delta\eta$,$\Delta\phi$ being the distances between the jet center and the constituent.

3.4.1 Jet System Kinematic Variables

As part of the analysis, there are a few observables of the jet system as a whole which are of interest in UPC studies: H_T, M_{jets} and y_{jets} which their definitions are given in the following equations:

$$H_T = \sum_i p_{T,i}\,, \tag{8}$$

$$M_{jets}^2 = \left(\sum_j E_j\right)^2 - \left(\sum_j \mathbf{p}_j\right)^2 , \tag{9}$$

$$y_{jets} = \frac{1}{2} ln \left(\frac{\sum_j E_j + p_{z,j}}{\sum_j E_j - p_{z,j}}\right) . \tag{10}$$

H_T is the scalar sum of the p_T of the jets, while M_{jets} and y_{jets} are the invariant mass and rapidity of the entire collection of jets.

3.5 Ultra Peripheral Collisions

Ultra Peripheral Processes are processes in which two nuclei pass each other with impact parameter larger than their radii sum. Instead, in these types of collisions the interaction is done via the photon flux surrounding the nuclei due to their acceleration. Using the EPA, it then follows that we can describe UPC as a set of processes in which there is a photon interacting either with the target nucleus or with a photon emitted by the target nucleus. In particular, we are interested in a subset of processes in which the final state is a di-jet. Under such constraints, we have a few different processes, which are labeled based on the type of interaction:

two-photon interaction ($\gamma\gamma$) In these interactions photons from both nuclei are interacting with each other. This can happen in 3 distinct topologies: direct-direct, in which there is a pure photon-photon interaction; direct-resolved, in which one photon dissociates into a $q\bar{q}$ pair that interacts with the photon; resolved-resolved, in which both photons dissociate and the remaining quarks interact.

photo-nuclear ($\gamma\mathbf{A}$) In this interaction the photon interacts with nucleons in the target nucleus. This can either result in nuclear breakup through electromagnetic dissociation, or not.

photo-diffractive ($\gamma\mathbb{P}$) In this interaction the photon does not interact with the nucleons directly. Instead, it interacts with the pomeron emitted by the target nucleus. The diffraction can then occur by either nucleus-generated pomeron (coherent diffraction) or a nucleon-generated pomeron (incoherent diffraction).

While these are good definitions for the initial state, things become more complicated as one tries to infer the initial state from the final state kinematics. Due to final state radiation (FSR) and hadronization, the final states in such collisions can vary greatly.

4 Deep Learning

Deep learning is a specific branch of machine learning dealing with neural architectures that contain many layers, often dozens or hundreds. This then allows them to learn hierarchical, high-level representations directly from raw data. Deep learning has exploded in recent years, with the advent of powerful GPUs, high performance computing clusters, and open-source software ecosystems such as PyTorch [18], which enable rapid experimentation and scalable

training across terabytes of data. In HEP, there have been many exciting breakthroughs,ranging from sophisticated jet-flavor taggers [19] and track-reconstruction networks [20] to generative models that accelerate detector simulation [21].

4.1 Neural network fundamentals

A neural network is a directed acyclic graph whose nodes (neurons) compute weighted sums of their inputs followed by a non-linear activation function σ. These activation functions are important in order to add non-linearity to the model, which allows it to model complex patterns. Usually, one can think of neural networks in terms of layers of processing, where the first layer in the model is usually referred to as the input layer, the last one as the output layer, and all the layers in-between are hidden-layers. Each layer operation (and output A) can be viewed by the following function (when considering fully connected networks):

$$A = \sigma(Wx + b)\,, \tag{11}$$

where x is a vector of inputs from previous neurons, W is the weight assigned to each neuron in the layer, and b is a bias one can introduce into the model. This linear calculation then passes through the activation function σ to generate the nonlinear output A.

4.2 Depth and hierarchical feature extraction

Depth refers to stacking many such layers. Each successive layer can learn higher-level features that combine lower-level patterns: In image tasks, early layers detect edges; later ones capture textures or object parts. This allows deep networks to model complex, non-linear relationships without explicitly engineering hand-crafted features. Theoretically, one can generate a single layer that can learn as well as any complex deep network (Universality approximation theorem), however, in practice with the compute power available, much better results are achieved by deep models.

4.3 Training via back propagation and stochastic gradient descent

Training a deep network means adjusting all weights so that the loss is minimized on the training data. A common way of doing that is via the backpropagation algorithm. This algorithm computes gradients of the loss with respect to every weight by applying the chain rule in reverse order through the network. These gradients are then used by an optimizer (commonly variants of stochastic gradient descent (SGD), e.g., Adam or RMSProp) to update weights.

4.4 Regularization, generalization, and overfitting

Deep models can contain millions of parameters; without constraints they can perfectly memorize training data (overfit). Regularization techniques help to mitigate this and allow the model to generalize better:

Dropout This technique disables connections between neurons during the training phase, thus breaking learned connections and forces the model to find alternative paths, which leads to better generalization.

L1 Regularization also called LASSO - Least Absolute Shrinkage and Selection Operator, L1 regularization refers to performing the typical L1 metric on all the weights in the model:

$$\mathcal{L}_{L1} = \sum_{L} \sum_{i} |w_{L,i}| \,, \tag{12}$$

where L is the layer number and i is the weight within the layer. This regularization promotes the use of small weights and drives insignificant weights to zero, thus making the model more sparse and less prone to over-fitting.

Weight Decay Similar to the previous one, weight decay is done by applying the L2 norm to the model weights. This causes the model to penalize high-value weights; however it will not actively promote sparsity in the model.

4.5 Architectural families

Dense Neural Network (DNN) DNNs are one of the most primitive but usually useful neural networks. In DNNs All the nodes of adjacent layers are connected to each other, allowing each node to adjust its weight based on all the inputs to the layer. This model type is still very useful when used in classifiers and is also used to some degree in more advanced models such as auto encoders.

Convolutional Neural Network (CNN) CNNs are a type of model which is mostly used with image data. The input image is applied sequentially on the various filters of the layer, in a manner similar to a sliding window. The values of the elements in the filter matrices evolve with the training of the model and produce a representation of information embedded in the image. The output of these filters is then flattened to a latent space which is pushed into a DNN classifier in order to produce the final logits.

Residual Neural Network (ResNet) ResNet is a technology built in 2015 on top of CNNs [22]. The main issue which can happen with both DNNs and CNNs is diminishing gradients in the calculation, in which the gradients zero out in a layer, thus not being able to propagate the calculation correctly over the graph. ResNet implements residual connections that bypass certain layers, and as a result can continue to propagate the information through the graph even if one layer zero out the gradients.

5 Previous Analysis

Previous analysis on the topic of di-jets in UPC has been done [23]. The ATLAS analysis, while looking for γA di-jet behavior with nuclear breakup, noticed that the cross section for di-jet events without nuclear breakup is much higher than anticipated. This increase in cross section is attributed to diffractive events, in which the photon interacts not with the nucleus itself but rather with a pomeron emitted by the probed nucleus.

6 Current Analysis

In this analysis we will examine data from 2018 Pb + Pb run. The aim of this analysis is to resolve ambiguities regarding the direction of the emitted photons in 0n0n events, as well as the ability to differentiate between different processes through the event topology. In addition, I propose to explore both basic and uncommon observables through the lens of the event topology, and aim to extract additional information on the event. The work already completed, which will be addressed in the upcoming chapter, concerns the use of MC to establish methods for photon direction discrimination, as well as an attempt to differentiate between γA, $\gamma\mathbb{P}$, and $\gamma\gamma$ samples. The planned work ahead, which will be described in the last chapter, will lay out the proposed research to improve upon these methods, as well as use them to extract new information from available data.

Part II
Analysis and Results

This chapter will focus on the analysis itself. It will outline the event selection and handling of backgrounds in data. Afterwards, the analysis of determining photon direction in 0n0n samples will be explored, as well as the generation of a machine learning framework to handle multiple questions relating to the topology of the event. Then results from different methods of the photon direction will be presented, as well as results of UPC process discrimination. This analysis follows all the same event selections and background subtractions listed in [23]. However, it focuses on events without nuclear breakup. The full event selection is detailed below:

7 Event Selection

The data used in the analysis is the 2018 Pb + Pb ATLAS data, given by the following triggers:

Trigger Chain	**Start Run**	**End Run**	**Prescale**	**Integrated Luminosity** $[nb^{-1}]$
0n0n Triggers				
HLT_j10_L1VZDC_A_C_TE5_VTE200	365498	366627	1.0	634.2
HLT_j10_0eta490_L1VZDC_A_C_TE5_VTE200	366691	367384	1.0	1083.6
0nXn Triggers				
HLT_j10_L1ZDC_XOR_TE5_VTE200	365498	366627	1.0	634.2
HLT_j10_0eta490_L1ZDC_XOR_TE5_VTE200	366691	367384	1.52	712.9

Table 1: Trigger chains and run statistics used in this analysis.

where the High Level Trigger (HLT) in all triggers is a jet above 10 GeV. For 0n0n, the level 1 trigger is an online requirement on ZDC which veto if any ZDC saw more than 5 TeV and less than 200 TeV. For 0nXn, the level 1 trigger requires exactly 1 ZDC with energy between 5 TeV and 200 TeV. In addition to the experimental data, MC is extensively used as well. The following MC samples were generated using Pythia8 [24] and then ran through the ATLAS reconstruction to produce reconstructed quantities to be used in the analysis.

- $\gamma A \rightarrow$ jets
- $\gamma \mathbb{P} \rightarrow$ jets (Both coherent and incoherent)
- $\gamma\gamma \rightarrow$ jets (direct-direct (DD), direct-resolved (DR,RD), and resolved-resolved (RR))

7.1 Offline Selections

Offline selections were applied equally to both Data and MC where applicable. This does not apply to ZDC as there was no ZDC simulation in the ATLAS reconstruction at the time of generating the MCs.

7.1.1 Track and Topocluster Selections

Track Selections Tracks are selected based on the pp MinBias working point, due to the low multiplicity in UPC collisions in addition to a 200 MeV p_T threshold requirement. The MinBias WP [25] requirements are:

- $|\eta| < 2.5$
- Minimum of 7 pixel and SCT clusters (12 are expected)
- Maximum of either one shared pixel cluster or two shared SCT clusters on the same layer
- Not more than two holes in the combined pixel and SCT detectors
- Not more than one hole in the pixel detector
- $|d_0^{BL}| < 2.0$ mm
- $|z_0^{BL} sin\theta| < 3.0$ mm

where a hole is an intersection of track trajectory with sensitive detector element without cluster, d_0^{BL} is the radial distance of the track with respect to the beam line ,and z_0^{BL} is the distance between the primary vertex and point in which d_0^{BL} is measured.

Topocluster Selections Topoclusters have the same minimum p_T requirement as tracks. In addition to that, there are the following requirements:

- Reconstructed clusters have to be outside the tile Gap ($1.0 < |\eta| < 1.7$).
- Cuts on cluster significance (for Tile ($\eta < 1.5$) and Hadronic End Cap + FCal ($\eta > 1.7$)).

7.1.2 UPC→ Di-jet Selections and Cuts

Vertex Cuts An event will be discarded if it has more than one reconstructed vertex. If there are more than 4 good tracks, it is required to have a vertex.

ZDC Energy Offline requirements Due to trigger turn on and post-run calibration, we enforce an additional offline cut for the ZDCs. As seen in Fig. 12, a cut at 40% of the value of the 1 neutron peak ($\sqrt{s_{NN}}/2$) is sufficient to set as an offline selection. Adjusting for 2018 1 neutron peak energy of 2.51 TeV, a threshold of 1 neutron at 1 TeV has been chosen to be conservative in the selection. For 0n0n samples, both ZDC require to have offline calibrated energy of < 1 TeV, while for the 0nXn sample, exactly one ZDC is required to have less than 1 TeV, while the other has to exceed that.

Gap Cuts A signature of photonuclear and photodiffractive events is the absence of particles (rapidity gap) in the direction of the nucleus which emitted the probing photon. This is due to the nucleus and photon being color neutral while the diffraction on the probed nucleus will generate color partons, that in turn will cascade into multiple hadrons in the final state. Usually, one would check for lack of particles on one side of the detector as a signature for this type of event, using a quantity known as a rapidity gap:

$$\Delta\eta_{\pm} = min|\pm 4.9 - \eta_i|| \,, \tag{13}$$

where η_i is the pseudorapidity of every track and cluster in the event, and ± 4.9 are the acceptance edges in ATLAS. However, stray particles may still appear near the edge of the detector. As a result, a much more robust quantity known as sum of gaps is used to account for this possibility. Sum of gaps ($\Sigma\Delta\eta$) is calculated by ordering all tracks and clusters in η and summing up all gaps if the gaps pass a threshold of 0.5. In the limit that there are no additional particles, this resolves back to edge gaps.

The cut is then made that the sum of gaps on either side of the detector has to be at least 2.0 for 0n0n events, while for 0nXn events the requirement is that the sum of gaps in the photon direction is larger than 2.5 units of pseudo-rapidity.

Jet requirements and Dijet Selection Each event is required to have exactly two reconstructed jets with calibrated energy of at least 15 GeV. In addition, in order to establish fiducial acceptance range in pseudo rapidity, the jet is required to have $|\eta| < 4.4$.

7.1.3 Non Physics Background

Splash Events Splash events are events in which the beam interacts with the LHC upstream of the IP, and thus sends particles into the IP. As the particles enter the IP only from the side which had the LHC-beam interaction, the resulting signature in the detector can mimic asymmetric events that will result in pseudo rapidity gaps mimicking real UPC events. However, in a real event there is a relation between the tracks and calorimeter clusters, while in splash events there will be a large deposit of energy clutsers without the appropriate number of tracks. Fig. 14 shows the cuts used to distinguish between splash events and real data in both 0n0n and 0nXn samples. As it is more likely the debris will also reach the ZDC, the 0nXn sample has higher contamination of splash events.

Out-of-Time Pileup Due to the high rate of collisions at the LHC, there is a possibility for 2 or more collisions to occur within a very short time window, such that tracks and energy deposits from one event will affect the recording of another. In particular, out-of-time (OOT) pileup events have a signature in which the energy deposited in the calorimeters by a previous event have not been collected in time, thus making the baseline charge for the next event higher than it should be. Events like this, following calorimeter calibration, will exhibit negative p_T and energy values. These events are then discarded by the following cuts:

- $\Sigma E_T^{FCAL} < -4$ GeV
- $-\Sigma P_T^{negCluster} > 5$ GeV
- $N_{negCluster} > 30$

Beam halo muons Beam halo muons occur in a similar way to splash events, but in this case the debris is stopped in LHC material, while energetic muons pass through the material, guided by the beam magnets and deposit their energy in the calorimeter. Because ATLAS was set up with a horizontal crossing angle for the beams, jets reconstructed from such

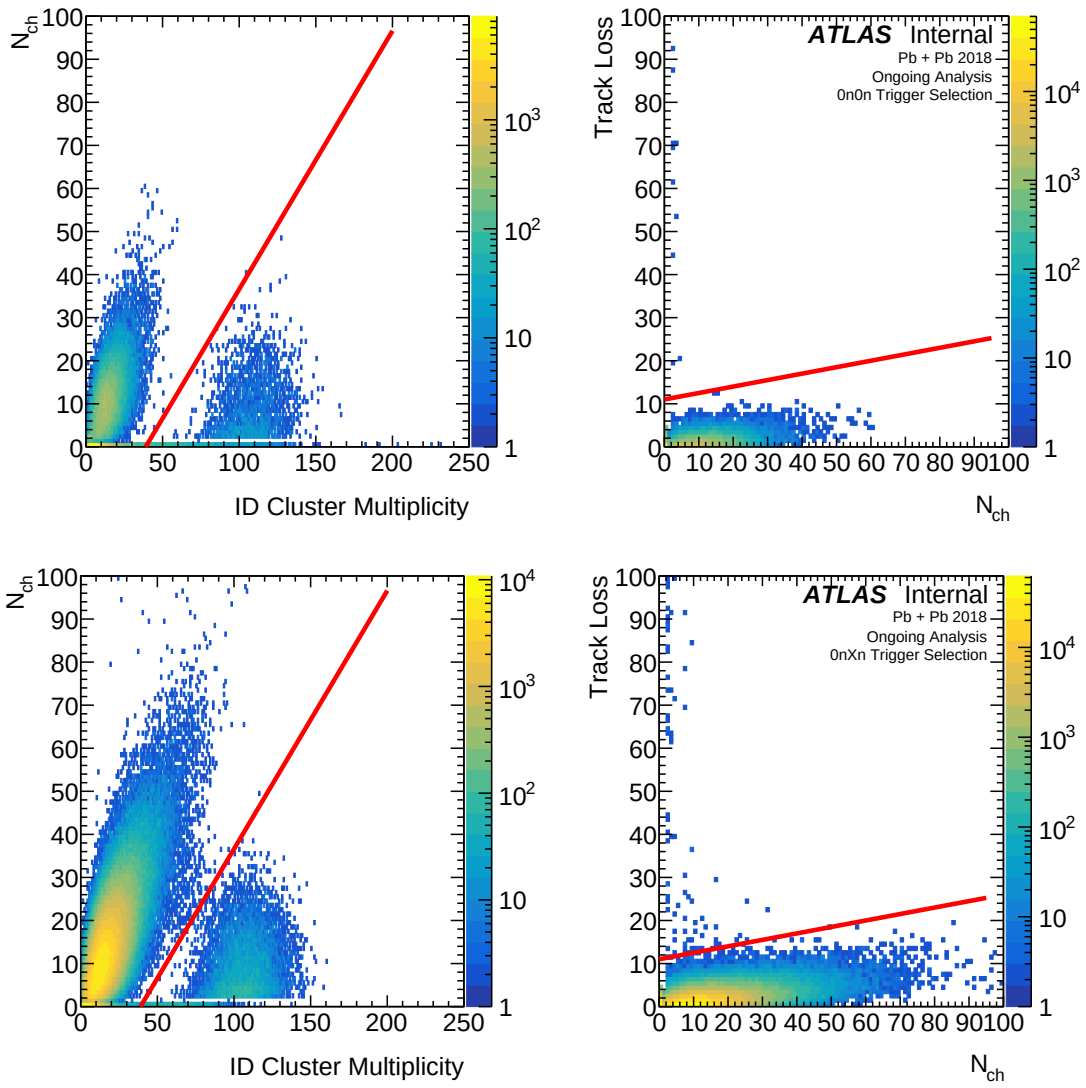


Figure 14: Representation of the cuts (in red lines) used to remove splash events. As expected, there is much higher contribution of splash events in the 0nXn (bottom two plots) sample due to the probability of neutral fragments reaching the ZDCs.

muons will be peaked at $\phi = 0, \pi$ while having no tracks pointing to them. Being guided by the beam magnets has another effect on those muons. As they were generated out of time with respect to the collision, their time distribution will be very broad with respect to the nominal jet times. Therefore, cutting on the jet time will provide good reduction for the beam halo muons. Following Fig. 16, a cut to deal with beam halo muons has been selected to be $t < -4.47$ or $t > 5.83$ ns for the 0nXn sample. The same selection remains for the 0n0n sample for comparative studies, although a more preferable cut towards the knee of the distribution will be studied in the future.

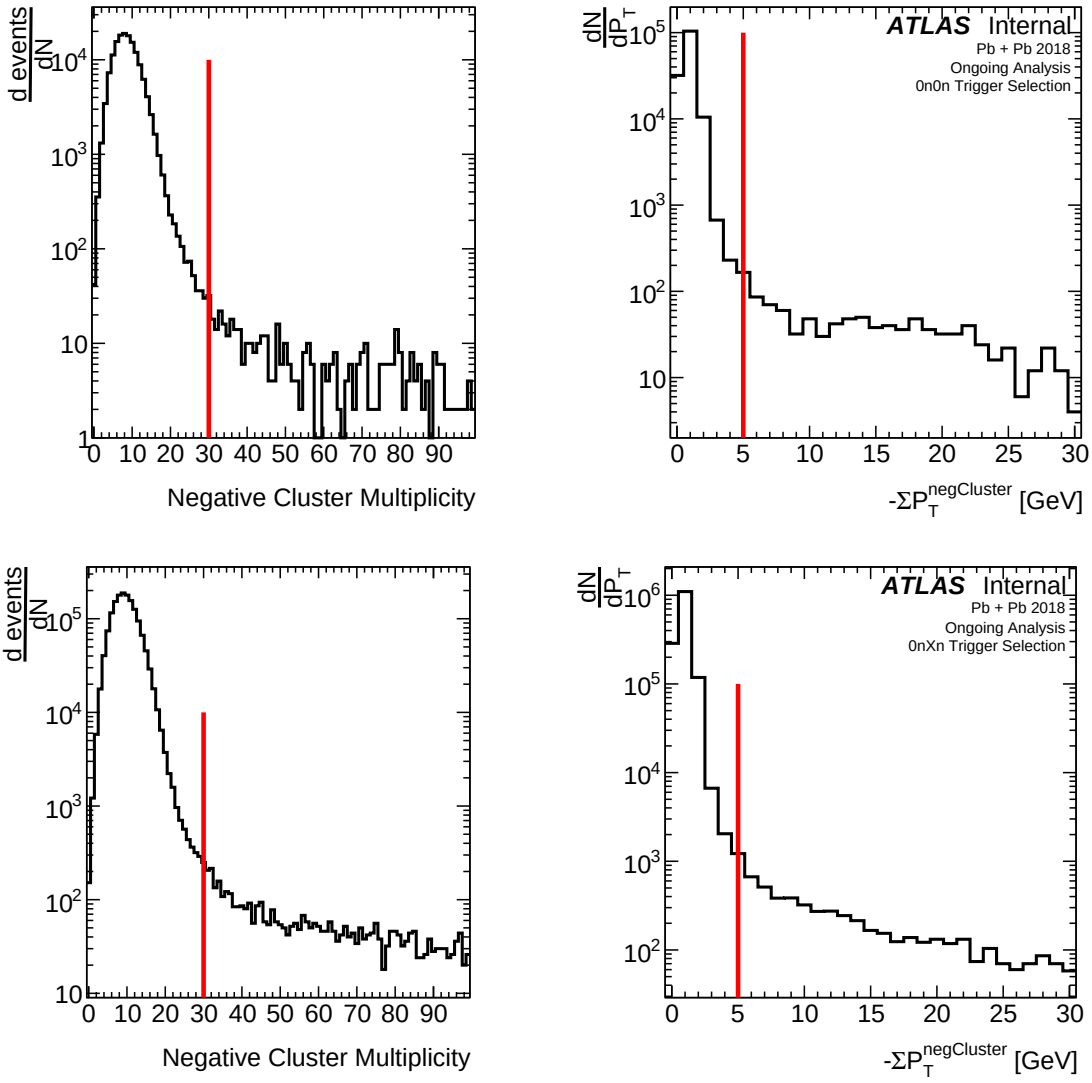


Figure 15: Representation of the cuts (in red lines) used to remove OOT pileup.

7.2 Selection Summary

Final selections and cuts are listed in Table 2 and Table 3. One can see that even within the sets of UPC triggers provided above, in order to choose the events of interest, there is a major reduction in available events for analysis.

8 Estimating Photon Direction

8.1 Estimating Photon Direction in UPC dijet events

When looking at UPC di-jet events, there are a few general properties which can be used to establish a sense of the probing photon direction. First, the di-jet rapidity for γA, will be produced in the probed nucleus center-of-mass, and therefore will be backwards to the probing

Selection/Cut	Prior	After	% Cut Retention	% Total Retention
Trigger Selection	—	3250685	N/A	N/A
Vertex Cut	3250685	2769180	85.19	85.19
ZDC Offline Selection	2769180	2543238	91.84	78.24
Gap Cuts	2543238	214196	8.42	6.59
Jet Selection	214196	36267	16.93	1.12
Splash Events	36267	34360	94.74	1.06
Out-of-Time Pileup	34360	33964	98.85	1.04
Beam Halo Muons	33964	17806	52.43	0.55

Table 2: The effect different selections and cuts have on the number of events for the 0n0n sample.

Selection/Cut	Prior	After	% Cut Retention	% Total Retention
Trigger Selection	—	20669012	N/A	N/A
Vertex Cut	20669012	18472620	89.37	89.37
ZDC Offline Selection	18472620	12239888	66.26	59.22
Gap Cuts	12239888	2684592	21.93	12.99
Jet Selection	2684592	697386	25.98	3.37
Splash Events	697386	686584	98.45	3.32
Out-of-Time Pileup	686584	681615	99.28	3.30
Beam Halo Muons	681615	647150	94.94	3.13

Table 3: The effect different selections and cuts have on the number of events for the 0nXn sample.

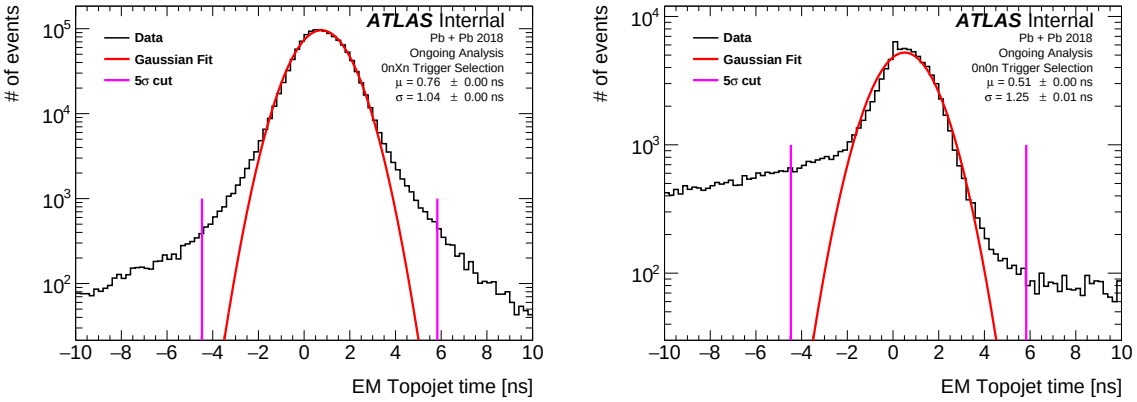


Figure 16: Plot of calibrated EM Topojets time with Gaussian fit. 5σ bands are shown in magenta to describe the cuts needed to reduce beam halo muons.

photon direction in the lab frame. Symmetric $\gamma\gamma$ contributions will have no side preference and therefore centered at zero rapidity, while asymmetric $\gamma\gamma$ will be forward with respect to the resolved photon. $\gamma\mathbb{P}$ will also be backward with respect to the photon direction, as can be seen in Fig. 17.

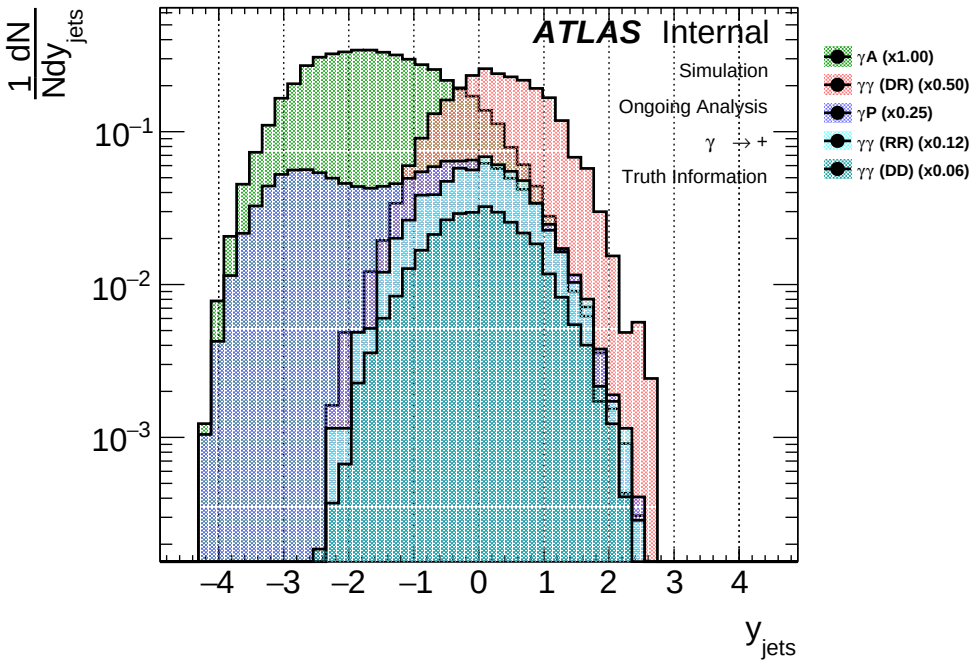


Figure 17: y_{jets} distribution for different MC samples. Nucleus emitting the probing γ is traveling towards forward rapidity. The $\gamma\mathbb{P}$ contribution includes 2 peaks: one closer to mid-rapidity, which stems from events where the nucleus underwent coherent diffraction, and one closer to backward rapidity, where the diffraction was stemmed from incoherent interaction.

Therefore, in samples with high fraction of photo-nuclear and photo-diffractive events as

produced by this analysis selections, a good first choice for photon direction is to choose the photon direction as opposite to the pseudo-rapidity in which the di-jet of the event was measured. We will refer to this method later as the "η method".

However, this method is inconsistent near mid-rapidity. In order to resolve that, one can check for other properties of UPC events. Given that we are looking at a photon-initiated event, the direction of the probing nucleus which emitted the photon should not have any color charge within it. In contrast, the probed nucleus which produced the di-jet, will have color-charged particles, which due to confinement, will generate new particles filling a band in η.

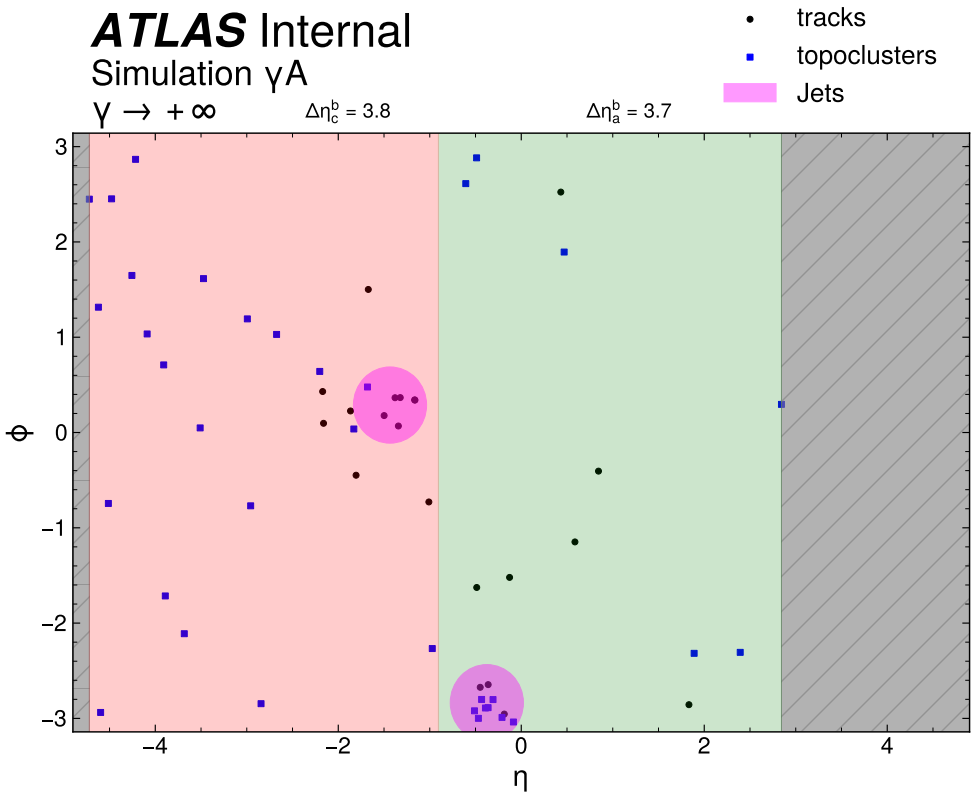


Figure 18: Event Display of a single γA event with the nucleus emitting the photon going towards positive η. The event is overlaid with the edge gaps and the particle bands. Edge gaps are in gray,A side band in green and C side band in red, with the reference point being the mean di-jet position in η. This specific example also shows the edge case in which detector acceptance can change the answer.

As can be seen in Fig. 18, identifying and selecting on that band can point into the direction of the probing photon. Therefore, this method, which will be named the "band method", decides on the photon direction by setting the di-jet pseudo-rapidity as an anchor point, and declaring two bands, one for each direction of the detector up to the last particle in acceptance in that side. The band of minimal length in η is then pointing to the direction of the photon.

It is important to clarify that this method only works if the particle trail is within the acceptance of the detector. Loss of acceptance can lead to misidentification. As a result, in

forward events, which are defined as events where the di-jet pseudo-rapidity is larger than 2.5, there will be a high chance of cutting the particle trail and making false predictions. To remedy that, such events will use the "η method", which will perform better for γA and $\gamma\mathbb{P}$ but will miss the few $\gamma\gamma$ that are forward enough.

(14)

9 Analysis Through Event Topology

9.1 Classification model based on event topology

Due to the immense phase space that exists with high multiplicities of tracks and clusters, it is extremely hard to search for kinematics and correlations. As discussed before, there are two main questions that are currently hard to answer when performing UPC analysis without breakup: namely, the interaction type and direction. In order to try and improve photon direction discrimination, as well as differentiate between γA, $\gamma\mathbb{P}$, and $\gamma\gamma$ contributions, one can search the phase space by using a neural network. Additionally, a neural network such as CNN or ResNet can also allow for the extraction of physics from the weights and activation maps post training. Below we will outline a generalized model for learning tasks in 0n0n UPC and the data pipeline associated with it.

As described above, convolutional neural networks are very appealing, as one can visually inspect the filters produced by the network to understand what part of an image the network is focusing on [26]. As mentioned in the introduction, since 2015 ResNets have been shown to outperform regular CNNs, while still having the ability to extract feature maps. As a result, this model will be composed of two parts:

Embedding Block This part of the network will have track and topocluster maps as input, in addition to event-level information as will be outlined below. The maps are processed through two separate ResNet-50 implementations, which then will be concatenated with the event data. The final layer in this implementation will embed the latent space of the events.

Decision Block The decision block will be selected based on the task at hand. For classification, a DNN will be used to differentiate between different process samples or between photon direction. Additional architectures can easily be implemented in order to perform other tasks.

9.2 Data preparation

To prepare the data for this type of search, each event produced two 77x77 pixel maps in $\eta - \phi$ space, one for tracks and one for topoclusters. The map consists of circles with centers at the position of the tracks/clusters, with an area corresponding to the track/cluster p_T. To make it easier to capture effects with respect to the jets, the leading jet was chosen as an anchor, and the map was rotated so that the leading jet is centered along ϕ .

In addition to the images, tabular data was also generated which contained the following observables:

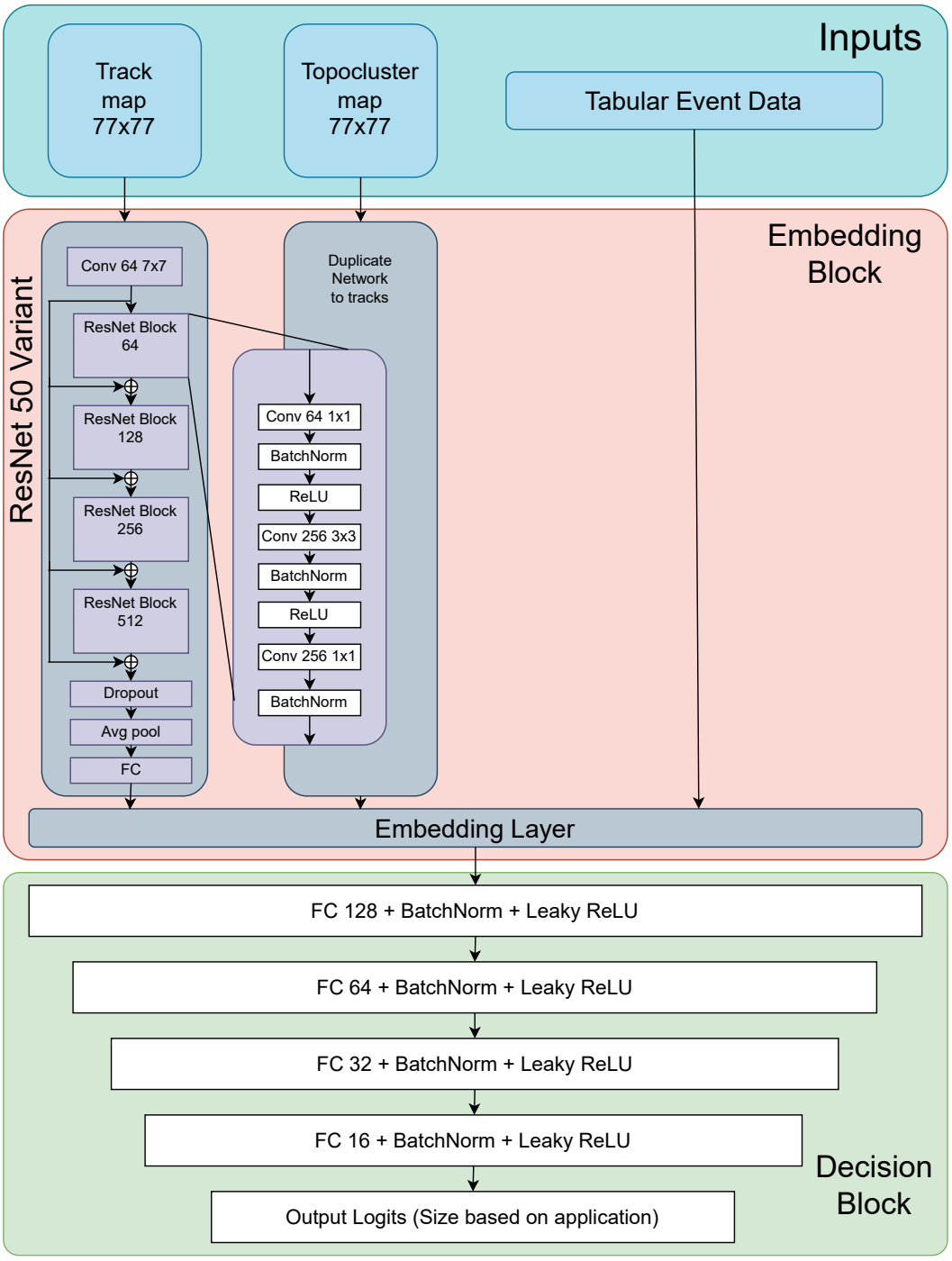


Figure 19: Diagram of the ML model. The track and topocluster maps receive the same treatment by two different instances of a ResNet 50 variant network, before being concatenated with tabular event data. Each of the ResNet blocks can generate activation maps of the filters associated with it.

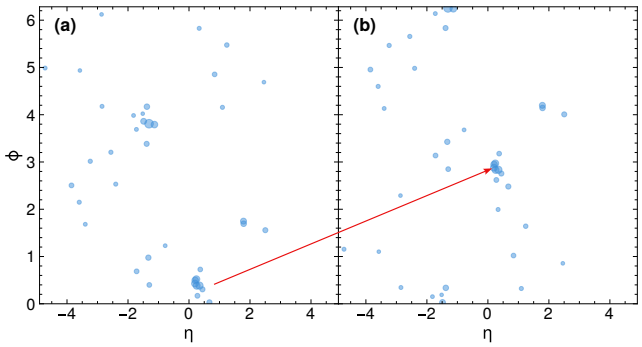


Figure 20: η-ϕ map of a single event, where both tracks and clusters are present. (a) Event map prior to rotation. (b) Post rotation.

- leading jet p_T ,η ,ϕ
- sub-leading jet p_T ,η ,ϕ
- H_T
- M_{jets}
- y_{jets}
- $\Delta\eta_A$, $\Delta\eta_C$
- $\Sigma\Delta\eta_A$, $\Sigma\Delta\eta_C$
- Estimated photon direction.

These values are normalized so that all values are in the region $[-1, 1]$ for numerical stability of the model. p_T, H_T and M_{jets} are converted to TeV units, while the various gaps and η are normalized to the detector acceptance range. ϕ is normalized to 2π .

9.3 Loss function

As established in section 4, the loss function is critical in order to establish the learning of the network. In this case, the network has to deal with highly imbalanced data, as a result of the analysis cuts and selections, and due to the complexity, has a high tendency to overfit the data. Therefore the loss function is set as a sum of different contributions to address these issues:

$$\mathcal{L} = \alpha(\mathcal{L}_{Dice} + \beta * \mathcal{L}_{OD} + \Gamma * \mathcal{L}_{CE} + \delta * \mathcal{L}_{L1}) \tag{15}$$

Dice Loss ($\mathcal{L}_{Dice}$) Dice loss is commonly used in medical ML application, such as tumor detection, has certain robustness to class imbalance [27]. The basic idea is to quantify the overlap between prediction and truth information per class of samples, using intersection and union. For binary classification, it is defined as:

$$\mathcal{L}_{Dice} = 1 - 2\sum_i \frac{y_i \cap \hat{y}_i}{y_i \cup \hat{y}_i}, \tag{16}$$

where y_i are the prediction per sample and $\hat{y}_i$ is the truth label for the ith sample. Its variation for the purpose of this network works as follows. The intersection is softened into a multiplication, while the union turns into addition, allowing the loss to be calculated with logits. This soft dice loss is seen in eq. 17

$$\mathcal{L}_{Dice} = 1 - 2\sum_{c}\frac{1}{N_c}\sum_{i}\frac{y_{i,c}\cdot\hat{y}_{i,c}}{y_{i,c}+\hat{y}_{i,c}+\epsilon}\,, \tag{17}$$

where c is the number of classes, i is the number of samples per batch and N_c is the number of samples from a specific class c. To properly describe the logits as probabilities, they are passed through the softmax function applied to all the classes for a given sample.

Off Diagonal Cost ($\mathcal{L}_{OD}$) The off diagonal cost is a term that helps to penalize misidentification more strongly than dice loss, and instead of worrying about the overlap, it penalizes deviations from full accuracy in proportion to the number of mismatched events.

$$\mathcal{L}_{OD} = \sum_{i} y_i(1-\hat{y}_i) \tag{18}$$

Cross Entropy Loss ($\mathcal{L}_{CE}$) Cross Entropy loss is a common classification loss in machine learning, and is widely used for classification tasks. It rewards good matching while penalizing deviations, as seen below:

$$\mathcal{L}_{CE} = -\sum_{i}\hat{y}_i * log(y_i) \tag{19}$$

L1 regularization ($\mathcal{L}_{L1}$) As explained in section 4, it adds an overall cost proportional to the absolute values of all the weights in the model, and effectively used to trim unneeded weights and paths to reduce over-fitting.

9.4 Training the model

For the training, we split the MC data into 70/30% split for training and testing, respectively. The training sample was then resampled to balance out the classes used during training, as a way to further control the CE portion of the loss function. The testing data remained imbalanced in order to see if the results of the network carry over to the imbalanced case. The optimizer selected for training is the Adam algorithm [28], which allows for more robustness. The hyper parameters used in the optimizer and loss function are listed in Table 4. This table lays out the hyperparameters used in the training of the model in order to reproduce the results shown later.

10 Results

10.1 Photon Direction

As described above, we check the photon direction against the base estimation that the photon direction will be opposite to the di-jet. We tested both the band method and a machine learning method based on the model outlined in section 9, where the classification head was set to distinguish only between photon sides. The inputs for the model are the same as discussed above, as well as the loss functions.

	Parameter	Value
Loss Function	α	$1e^4$
	β	50
	Γ	0.6
	δ	1
Optimizer	learning rate	$5e^{-7}$
	batch size	128
	weight decay	0.9

Table 4: Table describing the different hyper parameters used to train the model. These are listed for reproduction purposes.

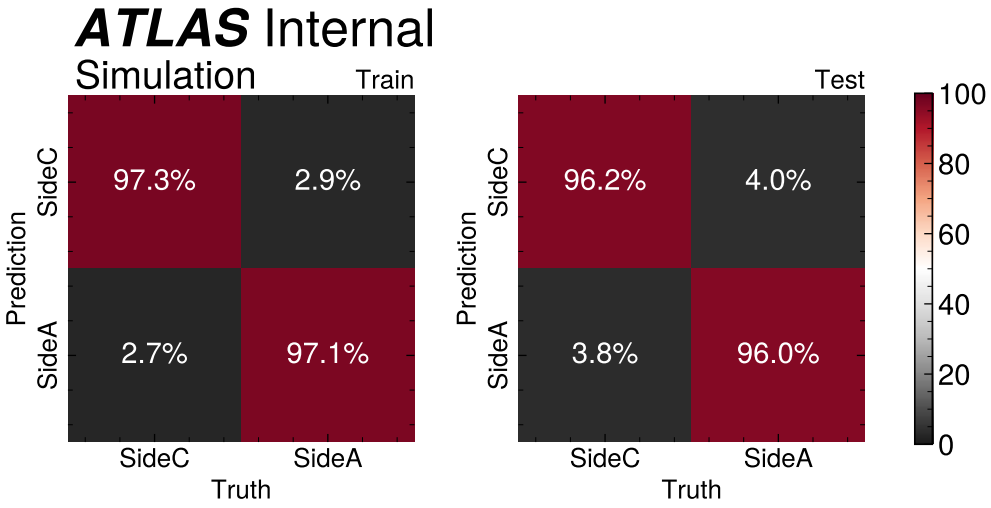


Figure 21: Confusion matrices for the training and testing samples for photon side discrimination. The plots show slight overfitting, but unbiased prediction on both samples.

The general performance results are summarized in Table 5. From it we can see that all methods are fairly unbiased with respect to a single detector side, as expected. In addition, both band and ML methods outperform the initial estimate, which we later see is due to the mid-rapidity di-jet events. These methods fix the issue of misclassifying the $\gamma\gamma$ samples which remain post event selection. These misidentifications are caused by the $\gamma\gamma$ direct-resolved contributions for which the di-jet rapidity is forward-biased instead of backward bias. But, as most of these events are located close to mid rapidity, they are handled well in both the band and ML methods. The distribution of accuracy with respect to H_T and y_{jets} was also tested. In Fig. 22 and 23, one can see the distributions of accuracy according to these observables.

In Fig. 23, one can clearly see the difficulty in assessing the direction at mid-rapidity for both γA and $\gamma\mathbb{P}$ while relying only on the di-jet position. The band method improves conditions in mid-rapidity, especially in γA. However, the ML model outperforms both methods and reaches an accuracy per bin of at least 90%. In addition, it can be seen that both algorithms outperform

Algorithm	0nXn†	γA	$\gamma\gamma$ Asym	$\gamma\gamma$ Sym	$\gamma\mathbb{P}$
η method	94.0	90.3	37.2	50.7	75.7
band method	94.6	93.4	79.9	49.4	91.0
ML method	97.3	99.7	78.0	43.8	95.7

† 0nXn "Truth" information was inferred from ZDC energy deposits.

Table 5: Table of sample accuracy (in %) for the 3 different algorithms for inferring γ direction. The $\gamma\gamma$ symmetric contribution shows the lack of bias to one of the sides. Uncertainties on these values are below 0.1%.

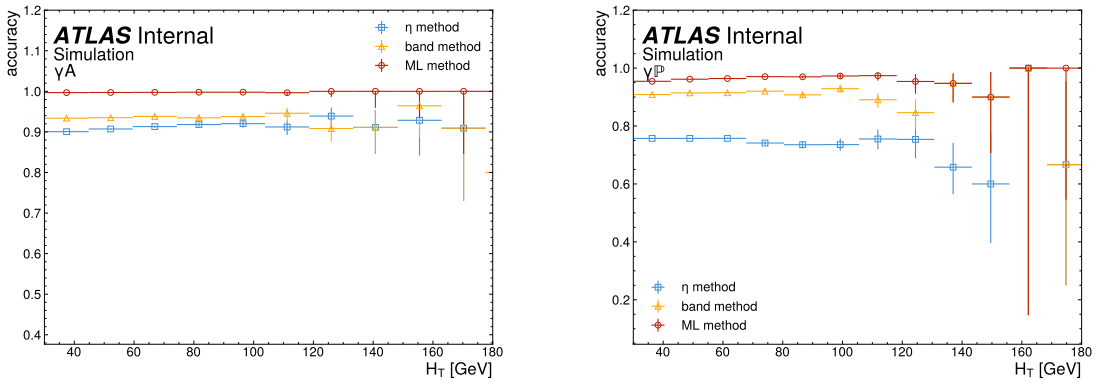


Figure 22: Accuracy plots for γA (left) and $\gamma\mathbb{P}$ (right) binned in H_T. Different colors assigned to the different methods used for the calculation.

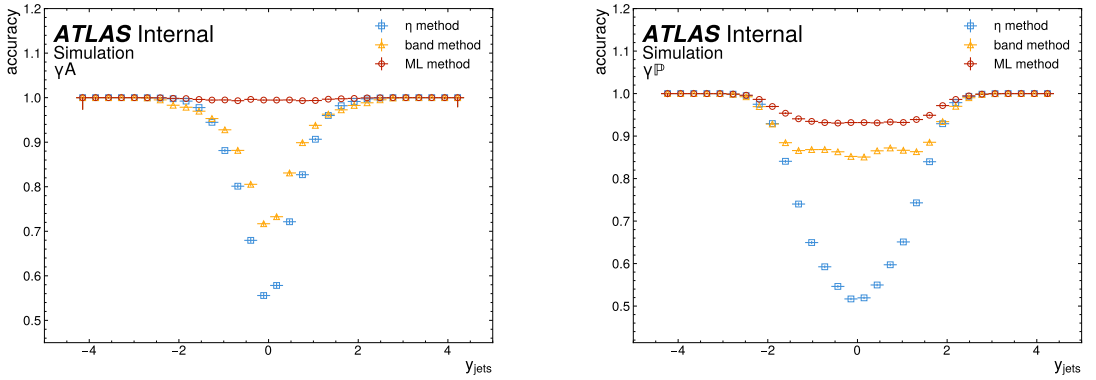


Figure 23: Accuracy plots for γA (left) and $\gamma\mathbb{P}$ (right) binned in y_{jets}. Different colors assigned to the different methods used for the calculation.

the η method at low H_T values, and barring statistical uncertainties, continue to exhibit better performance at high H_T bins, showing it is applicable over a wide range of the phase space.

Finally, while the models were prepared with the 0n0n case in mind, given their performance, a check against data in 0nXn is performed, where the ZDC provides the "truth" information.

In Fig. 24, One can see that the results in H_T and y_{jets} are comparable to the MC.

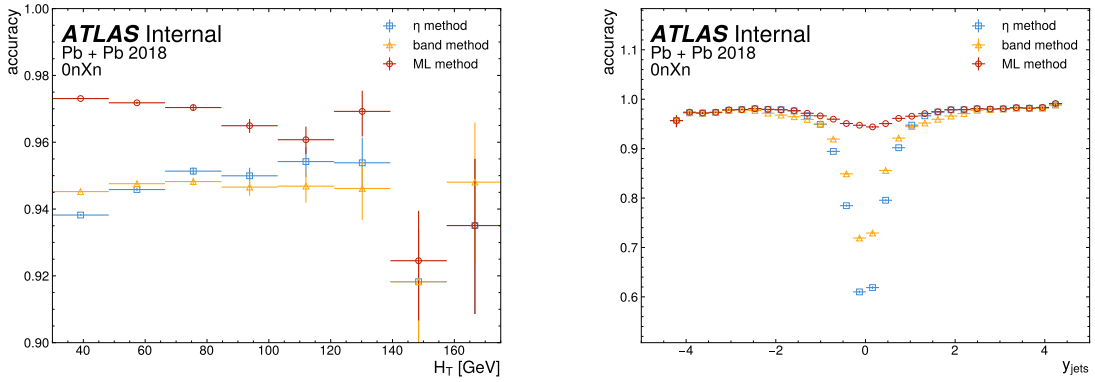


Figure 24: Accuracy plots for 0nXn binned in H_T (left) and y_{jets} (right). "Truth" information defined by ZDC X direction.

10.2 UPC Process Discrimination

10.3 Sample Discrimination

In the same way as the machine learning approach for the photon direction, by adjusting the target criteria and the decision block, the same model can be used to differentiate between different processes. A few different methods have been tested: Multi-Class discrimination and more general binary arch-type discrimination, which will be detailed below.

10.3.1 Multi Process Discrimination

Given the MC samples of multiple initial states, as described in section 7, all of them could be placed into the model at once in order to make granular separation. The decision block structure for this task is outlined in Fig. 19. The latent space from the embedding block is passed through a DNN classifier, which is terminated by a 5-neuron layer which provides the logits related to the probability of the sample being any of the 5 processes (classes in ML jargon). These logits then go through a softmax activation function and are compared to a hot-encoded target value. The results are being shown in Fig. 25.

As can be seen, the accuracy, as indicated by the numbers along the diagonal, are high enough to disprove combinatorial success. However, looking at both the training and testing samples, it is clear there is over-fitting in place. Further, it can be seen by the position of high inaccuracy off-diagonal elements that the model is struggling in distinguishing between samples within the $\gamma\gamma$ branch of processes, most likely due to low statistics in the number of events from each group. In addition, as we expect, in broad terms, for $\gamma\mathbb{P}$ to be between $\gamma\gamma$ and γA with respect to to the momentum transfer to the nucleus, we also see the inaccuracies for $\gamma\mathbb{P}$ being shifted between γA and $\gamma\gamma$ resolved-resolved predictions.

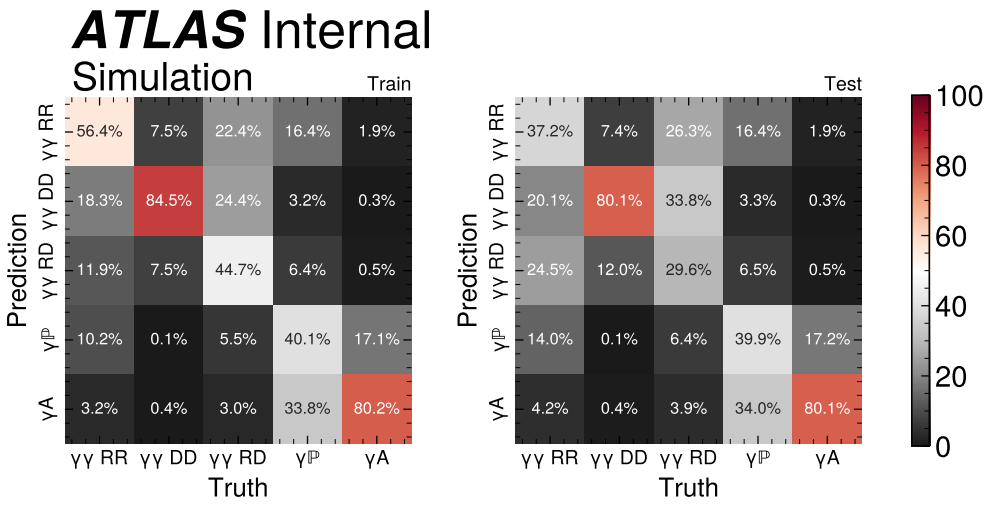


Figure 25: Confusion matrices for the training and testing samples for multi-class classification. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions. There is some block structure showing separation of $\gamma\gamma$ from γA , $\gamma\mathbb{P}$ contributions, while there is some overlap between $\gamma\mathbb{P}$ and $\gamma\gamma$RR.

10.3.2 Binary Process Discrimination

Due to the results above, we present below the attempt to first make discrimination between $\gamma\gamma$ and γA processes, and the effect it has on the $\gamma\mathbb{P}$ sample. The various samples from all the $\gamma\gamma$ processes were consolidated under a single class, while γA was designated as the 2nd class. In addition, the decision block was changed to provide two-class logit outputs. The results of this classification are presented in Fig. 26.

The over-fitting is low compared to the previous model, and the accuracy is much higher. In addition,two other models, used to differentiate γA and $\gamma\mathbb{P}$ and $\gamma\gamma$, $\gamma\mathbb{P}$ samples were trained, with the results given in Fig. 27 , 28 . As expected from the multi-class model, and from our understanding of the similarities between the final states of both samples, the results are worse than in the $\gamma\gamma$, γA case. The current stopping point for the training samples was a balanced accuracy across samples, although there is still work to be done, if, due to cross sections a bias towards higher efficiency of the dominant samples will be preferred.

Finally, all models (including photon direction) were applied to both 0n0n and 0nXn 2018 data samples, and were then used to reconstruct from data the y_jets distribution, for comparison with Fig. 17. The decision process on the sample classification can be seen in Fig. 29, and is informed by the results we see from the multi-class classification in Fig. 25. Due to the block matrix structure, the $\gamma\gamma - \gamma A$ classification is made, and then the comparison of each to $\gamma\mathbb{P}$ is made.

The results are shown in Fig. 30. It is important to note that there are currently no uncertainties associated with the models that were propagated to the figure, and Fig. 17 was not corrected for cross sections between samples and between $\gamma\mathbb{P}$ different contributions.

As can be seen in Fig. 30, the peaks of γA and $\gamma\mathbb{P}$ match quite well with MC. in particular,

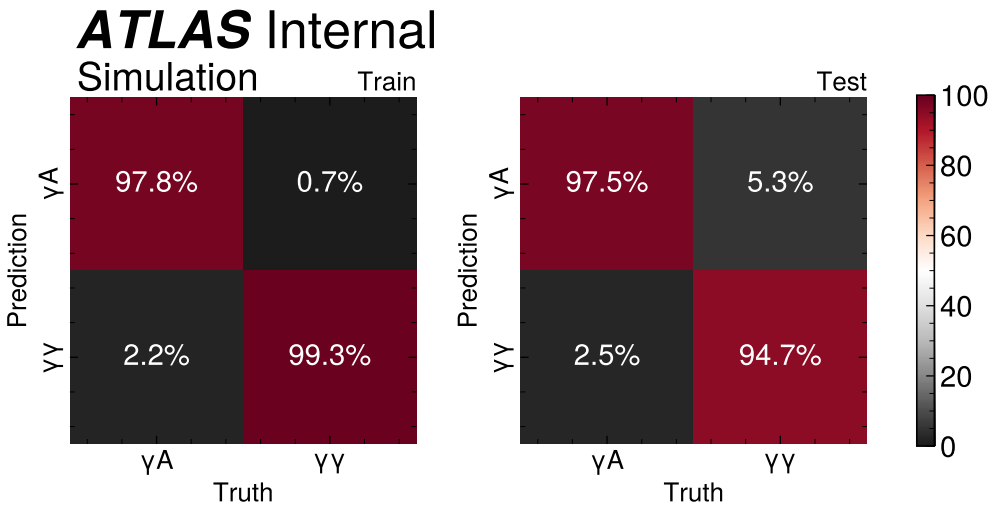


Figure 26: Confusion matrices for the training and testing samples for binary $\gamma\gamma$, γA where all $\gamma\gamma$ processes were labeled as one. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions.

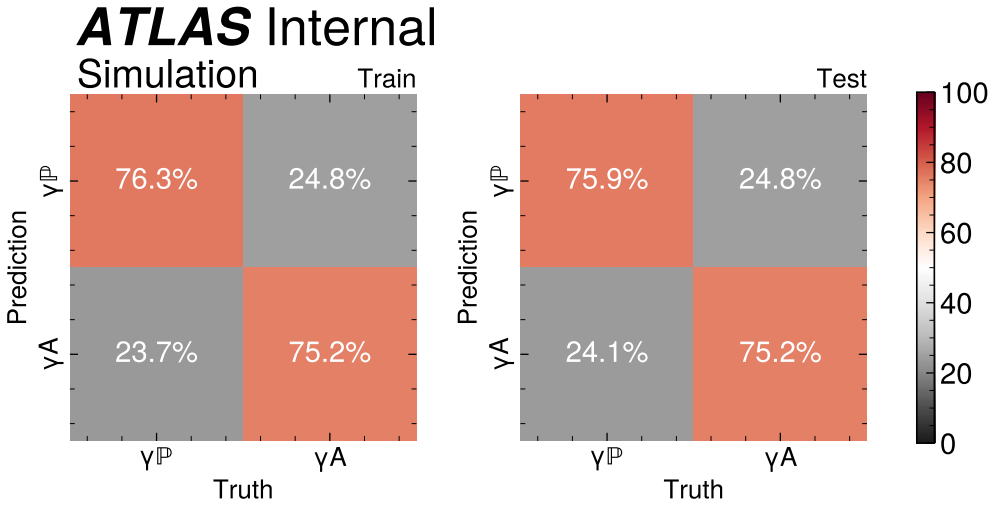


Figure 27: Confusion matrices for the training and testing samples for γA and $\gamma\mathbb{P}$ processes. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions.

it is interesting to see that the models suggest that in 0n0n there is much more photo-diffractive contribution in the form of a coherent process, giving jets closer to mid-rapidity, but in the 0nXn sample, almost all the contribution comes from incoherent diffraction, as expected given the sample is selected by nuclear breakup. This plot is an encouraging first preliminary result that there is a measurable difference between γA and γP contributions in the final states, which

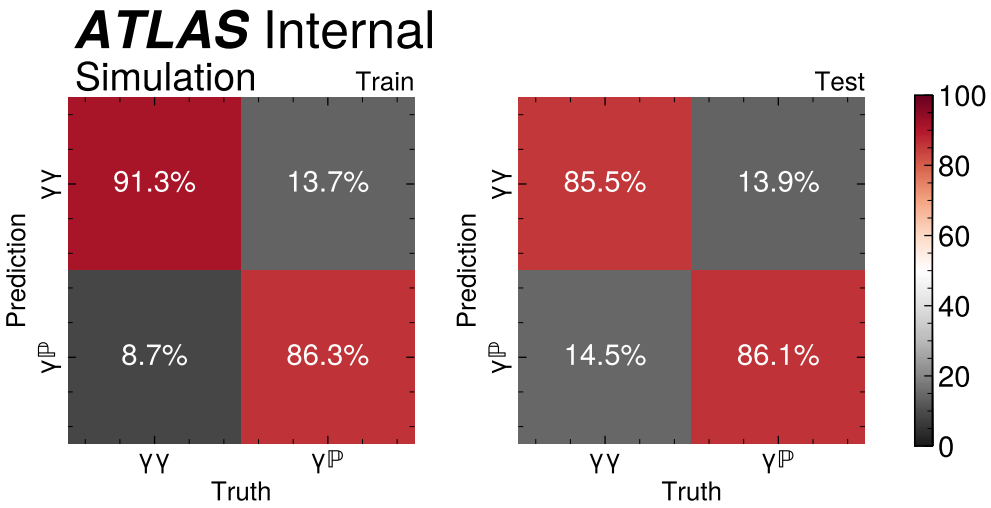


Figure 28: Confusion matrices for the training and testing samples for $\gamma\gamma$ and $\gamma\mathbb{P}$ processes. Percentages sum along the columns to 100%, thus showing the distribution of truth with respect to different predictions.

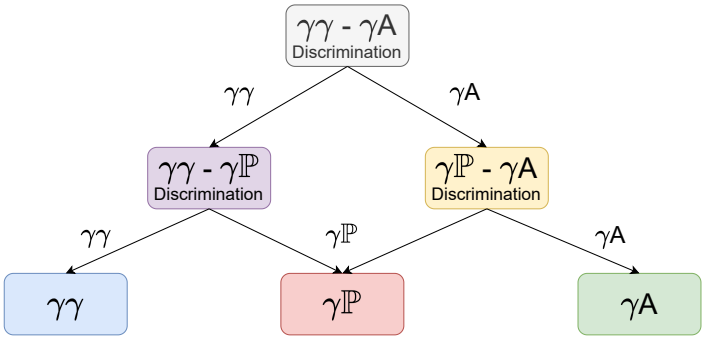


Figure 29: Diagram of the decision process for final sample classification based on the results of the different models.

can be explored and used to make better selections for future measurements.

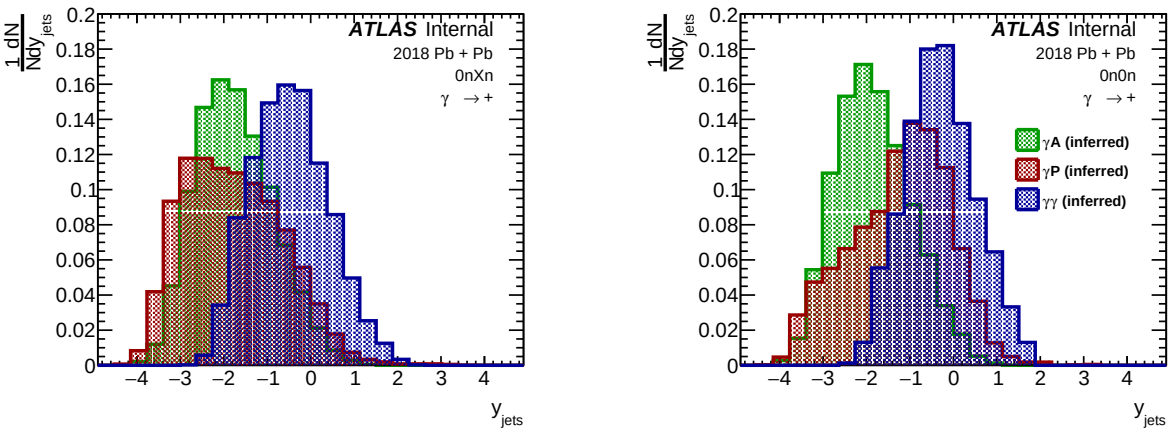


Figure 30: Plots of y_{jets} for 0nXn (Left) and 0n0n (Right), separating out the different contributions according to the ML models. Events were selected based on the photon direction model such that the nucleus emitting the photon goes in positive rapidity.

Part III
Follow up research

This chapter will cover the follow-up research to be done, using the results provided above. Following the results presented above, the next steps in the analysis will be as follows:

Evaluation of systematics The results given above for the accuracies have only statistical uncertainties applied to them. Therefore, in order to have good control over the results, method-dependent systematic uncertainties and closure tests evaluated.

Extracting physics from the ML model One of the reasons for creating the ML model as built was to extract new observables and identify differences between the samples provided. As mentioned in section 4, the inclusion of the ResNet models allows us to look into the decision making process of the model, and analyze the prominent filters in a single event. Given the results of the model, it is clear there is a latent space in which it is possible to distinguish between different photon induced processes , resulting in di-jets. Therefore, identifying these observables could lead to better understanding of such processes.

Processing Run 3 data in order to increase statistics for future measurements As mentioned, The results previously used only the 2018 Pb + Pb data. In addition, the reduction in number of events due to event selection, as seen in section 7, means that there is a lack of statistics to achieve precision. Thus, one of the important tasks is to increase the number of events available by going through and processing Run 3 UPC di-jet events, an effort which will include participating in the jet calibration process in ATLAS for all the Run 3 datasets, as well as other tasks that may arise in getting the data to the position in which the same analysis can be run on.

Validation of the $\gamma\mathbb{P}$ MC sample While we use the $\gamma\mathbb{P}$ MC in this analysis, there are some open questions regarding its handling of neutrals, etc. In addition, there is currently no other MC to compare against, which has been processed through the entire ATLAS analysis chain. Therefore another task is to conduct a detailed validation of the MC.

In addition to all of these tasks, There is also work being done on the ZDC (The sub detector mentioned in section 3.3). As part of the ZDC team and an expert on ZDC TDAQ infrastructure, There is additional work that will carry on with respect to the last runs of Run 3 and preparation for Run 4. In contrast to the amount of information presented here, work on the ZDC takes considerable amount of time, but the result is better data taking that can be used for future measurements.

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